Phase space mixing in the presence of self-gravity: spherically symmetric Vlasov-Poisson evolutions from initial data out of equilibrium

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[(Erik) Provisional title and abstract; to be rewritten when sections VI and VII are final.] These notes are devoted to the numerical study of a spherically symmetric, collisionless kinetic gas which is trapped in an external central potential, whose self-gravity is taken into account, and whose initial distribution function is not a steady state. In the absence of self-gravity, such a gas relaxes in time to a stationary state through phase space mixing. We analyze how this relaxation is modified by the self-gravity of the gas, for the reduced model in which all the gas particles have the same magnitude of the angular momentum. Our study is based on a particle-in-cell code, on a numerical solution of the linearized equations in action-angle variables, and on a spectral criterion which characterizes the existence of oscillating modes. For initial data close to a self-consistent steady state, the deviation from that state is damped unless the steady state admits an oscillating mode, which happens above a threshold mass that is accurately predicted by the spectral criterion, and at late times the self-gravity slows down the algebraic decay of the observables. For initial data far from equilibrium, the gas mixes when its self-gravity is weak and collapses into a self-bound core which does not mix when its self-gravity exceeds a critical value.

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I. INTRODUCTION

Collisionless kinetic gases provide the standard description for systems consisting of a large number of particles which interact with each other only through the mean field they generate. Prominent examples are the stars in a galaxy or in a globular cluster, and the dark matter particles in a galactic halo [1]. The state of such a gas at time t is described by its one-particle distribution function (DF) $f(t, x, p)$, a nonnegative function on the one-particle phase space which determines the number of gas particles in a small phase space volume about the point (x, p), and whose evolution is governed by the Vlasov equation.[1] In the Newtonian gravitational case, the Vlasov equation is coupled to Poisson's equation for the gravitational potential generated by the gas, which yields the Vlasov-Poisson system, see Ref. [2] for a review of its mathematical properties. Its general relativistic counterpart is the Vlasov-Einstein system [3].

It is instructive to compare the dynamics of a self-gravitating collisionless gas with the one of a pressureless fluid (dust), which is obtained in the limit in which the velocity dispersion of the gas vanishes. The gravitational collapse of a spherical dust cloud leads to the formation of a singularity in finite time at which the density diverges, both in the Newtonian theory and in general relativity [4]; in the latter case, part of this singularity may even be visible to distant observers [5, 6]. In contrast to this, the velocity dispersion of a kinetic gas prevents the particles from focusing. In fact, the Vlasov-Poisson system admits global classical solutions for general initial data [7–9], while for the spherically symmetric Vlasov-Einstein system global solutions exist for small initial data [10] and sufficiently concentrated data collapse to a black hole [11]. Therefore, for a Newtonian collisionless gas the relevant question is not whether or not a singularity forms, but rather how the gas behaves for large times and, in particular, whether or not it relaxes to a stationary state.

A fundamental mechanism leading to such a relaxation is phase space mixing [12, 13]. To explain it, consider first a gas whose self-gravity is neglected and which is trapped in the potential well of an external potential. If the one-particle Hamiltonian is integrable, each gas particle is confined to an invariant torus, and in terms of action-angle variables (Q, J) the solution of the Vlasov equation is $f(t, Q, J) = F(Q - \Omega(J)t, J)$, where F describes the initial DF and $\Omega(J)$ denotes the frequencies associated with the torus labeled by J. Since these frequencies vary from one torus to the other, the DF is stretched along the angle directions and develops structures on finer and finer scales as time increases. As a consequence, f does not converge pointwise as $t \to \infty$. However, the macroscopic observables which

[1] The Vlasov equation is also often called the collisionless Boltzmann equation in the literature.

are obtained by integrating f against a test function do converge, and their limit is the one associated with the average of the initial DF over the angle variables, which depends only on integrals of motion. Note that this holds for any initial DF, not only for small perturbations of a steady state. A mathematically precise formulation of this statement and its proof for rotationally symmetric potentials in $d = 1, 2, 3$ space dimensions satisfying a suitable non-degeneracy condition were given by Rioseco and Sarbach [14] (see also Refs. [15, 16]). The results in Ref. [14] include the case of a spherically symmetric gas configuration in which all the gas particles have the same value of the total angular momentum, which behaves like an effective one-dimensional system, and they explain the relaxation to a virialized state which had been observed in the numerical simulations of Ref. [17]. The final state is thus determined by the initial DF through its average over the angle variables; for a gas in an axisymmetric potential with two potential wells along the symmetry axis, in which the motion is integrable, this prediction has recently been compared with N-particle simulations in Ref. [18]. For the analogous phenomenon in the exterior of a black hole we refer the reader to Refs. [19, 20], and for estimates on the rate at which the observables converge to Refs. [21–23].

The generalization to the self-gravitating case, which was left open in Ref. [14], is considerably more involved, since the potential now depends on the DF, such that a change of the latter induces a change of the potential which, in turn, acts back on the gas. In plasma physics, where the particles interact through their mean electric field, the corresponding collective effect has been known since the work of Landau [24]: for a gas of charged particles close to a spatially homogeneous equilibrium the electric field decays in time while the DF undergoes mixing. This phenomenon, known as Landau damping, has been established rigorously at the nonlinear level by Mouhot and Villani [25]. The gravitational case differs from the electromagnetic one in two important aspects. First, the interaction is attractive, such that the collective response of the gas tends to reinforce a perturbation instead of screening it. Second, a self-gravitating gas configuration of finite mass does not possess spatially homogeneous equilibria. Instead, its steady states are spatially inhomogeneous, the gas particles are trapped in the potential well, and the frequencies of their radial motion lie in an interval which, in the situations considered in these notes, is bounded and bounded away from zero. For this reason, a collective oscillation whose frequency lies below this interval is not in resonance with any gas particle and hence cannot be damped by mixing.

Two situations have been analyzed in the gravitational case. The first one concerns small perturbations of a self-consistent steady state. Hadžić, Rein and Straub [26] (see also Refs. [27, 28]) showed that the essential spectrum of the operator governing the linearized dynamics about a large class of spherically symmetric steady states is determined by the radial periods of the gas particles, and, using a version of the Birman-Schwinger principle which was first applied to this problem by Mathur [29], they derived a criterion for the existence of an eigenvalue below this spectrum, which corresponds to an oscillating mode that is not damped. Related criteria and estimates for the number of such modes have been given in Ref. [30]. For the reduced system in which all the gas particles have the same magnitude of the angular momentum and move around a central point mass, Hadžić, Rein, Schrecker and Straub [31] proved the following dichotomy for steady states of sufficiently small mass whose DF is proportional to $(E_0 - E)^k$ for energies E below the cutoff energy E_0: the solutions of the linearized system are damped if $k > 1$, whereas an oscillating mode exists if $1/2 < k \leq 1$. Quantitative decay estimates for the linearized system have been obtained in Refs. [23, 32], and particle-in-cell simulations show that, depending on the steady state, its perturbations either oscillate without damping or decay [33, 34]. The second situation concerns initial data which are not a perturbation of a steady state of the gas. For a self-gravitating gas around a point mass, Chaturvedi and Luk [35] proved that small, spherically symmetric initial data which are supported on bounded orbits, with angular momenta in an interval bounded away from zero, undergo phase space mixing up to a time of the order of $\epsilon^{-1}(\log \epsilon^{-1})^{-2}$, where ϵ measures the size of the data, while the global existence and the asymptotic behavior of these solutions remain open. Finally, when the initial data are far from any steady state and the self-gravity is strong, the gas undergoes a violent relaxation [13] toward a quasi-stationary state, and the prediction of this state from the initial data is understood only in special cases, see Ref. [36] for a review.

In these notes, we study numerically the relaxation through phase space mixing of a spherically symmetric collisionless gas which is trapped in an external central potential, whose self-gravity is taken into account, and whose initial DF is not a steady state. We consider the reduced model in which all the gas particles have the same magnitude $L_0 > 0$ of the angular momentum, which has a two-dimensional phase space and is the setting considered in Refs. [14, 31]. This model goes back to the numerical experiments of Hénon with spherical shells (see Ref. [37] and references therein) and has been used in the N-body simulations of Ref. [38]. For the external potential we choose the isochrone potential, for which the action-angle variables are known explicitly, and the potential of a point mass, which is the one considered in Refs. [31, 35]. We address three questions: how fast the macroscopic observables relax and what determines the rate, what can prevent the mixing from being completed, and which state the gas reaches at late times. Our analysis is based on three tools: (i) a particle-in-cell (PIC) code which solves the nonlinear Vlasov-Poisson system, (ii) a numerical solution of the equations linearized about a self-consistent steady state, formulated in the action-angle variables of the latter, which describes initial data close to that steady state, and (iii) the numerical computation of the largest eigenvalue $\lambda(\omega)$ of the operator of Mathur, which measures the gain of the self-consistent

loop in which a perturbation of the potential with frequency ω induces a perturbation of the density which, in turn, generates a potential. An oscillating mode with frequency below the frequency band of the gas particles exists if and only if the value λ_{edge} of this gain at the lower edge of the band is larger than one. This is the criterion of Refs. [26, 28, 29]; the quantity λ_{edge} corresponds to the number which is denoted by M in Ref. [26] and by μ_* in Ref. [28].

The main results of these notes can be summarized as follows. First, in the absence of self-gravity the observables relax algebraically, with an exponent which is determined by the behavior of the initial DF at the boundary of its support: a DF which vanishes like $(J_{max} - J)^k$ at the outer edge of its support gives rise to a tail proportional to $t^{-(k+1)}$, and the PIC code reproduces the exact solution of this problem. Second, for initial data close to a self-consistent steady state, the self-gravity of the deviation from that state modifies the relaxation in two ways. If $\lambda_{edge} < 1$, the deviation is damped, first exponentially, as in Landau damping, and later algebraically; at late times, or for larger masses, the self-gravity reduces the exponent of the algebraic decay from $k+1$ to k, which is the exponent of the decay estimates of Ref. [23] for perturbations that vanish at the edge like the derivative of the DF of the steady state. If $\lambda_{edge} > 1$, the mixing is not completed, since the part of the deviation which is carried by the oscillating mode persists. For the families of steady states we consider, this happens above a threshold mass, and the frequency predicted by the equation $\lambda(\omega) = 1$ agrees with the one measured in the simulations. Third, the oscillating modes which lie close to the edge of the band can only be sustained with a small amplitude: above a critical amplitude, which we estimate by means of a pendulum model, the gas particles near the edge of the steady state are trapped by the mode. That the amplitude of such modes is limited was observed by Weinberg [39] in simulations of the one-dimensional slab; the pendulum model provides an explicit estimate for the amplitude at which this limitation sets in. Fourth, for initial data far from any steady state, in which the gas is concentrated about one orbital phase, the outcome is determined by the ratio μ between the bounce frequency of the gas particles in the potential well of the gas and the spread of their orbital frequencies: for small values of μ the gas mixes, whereas above a critical value of μ, which lies between 0.3 and 0.5 in our simulations, it collapses into a self-bound core which does not mix. [(Erik) Preliminary result from the runs B0–B5 of the demo document. To be consolidated in section VII: more masses, the dependence on the widths of the initial DF, convergence, and the final state.]

Particle methods have a long tradition in the numerical study of spherically symmetric collisionless systems, both in the Newtonian case [33, 34] and in the relativistic one, where they have been used to analyze the collapse of star clusters and the critical behavior at the threshold of black hole formation [40–44]; see Refs. [45–47] for introductions to the PIC method, Ref. [48] for a higher-order implementation in the context of plasma physics, and Ref. [49] for a convergence proof of a PIC scheme for the spherically symmetric Vlasov-Einstein system. Measuring the relaxation of the observables with such a method is delicate, because the signal has to be separated from the noise which is due to the finite number of computational particles. In these notes we address this problem by placing the particles on a regular lattice in action-angle variables, by encoding a deviation from a steady state in the weights of the particles instead of their positions (as in Refs. [50, 51]), and by subtracting a reference run without this deviation. A loading of the particles which reduces the initial fluctuations (a quiet start) was already used for systems of spherical shells in Ref. [37]. We also show that the observables need to be computed with the action-angle variables belonging to the total potential, since those of the external potential alone leave a time-independent residue which could be mistaken for an incomplete mixing. For initial data far from equilibrium there is no fixed potential with respect to which the gas relaxes, and we use the action-angle variables of the instantaneous potential.

The remainder of these notes is organized as follows. In section II we formulate the Vlasov-Poisson system with an external potential in spherical symmetry and the reduced model with fixed angular momentum, and we introduce the action-angle variables and the observables used throughout these notes. In section III we review the mixing in the absence of self-gravity, derive the linearized equations about a steady state and state the criterion for the existence of oscillating modes. Our numerical methods are described in section IV, and their validation is presented in section V. Section VI is devoted to initial data close to a self-consistent steady state, and section VII to initial data far from equilibrium. Conclusions and an outlook, including the extension to a distribution of angular momenta, are given in section VIII. Technical details regarding the numerical computation of the action-angle variables are included in Appendix A.

II. THE VLASOV-POISSON SYSTEM IN SPHERICAL SYMMETRY

In this section we formulate the Vlasov-Poisson system with an external potential and reduce it to spherical symmetry, first for a gas whose particles have arbitrary angular momenta and then for the reduced model in which all of them have the same magnitude of the angular momentum. Next, we introduce the dimensionless variables used in our numerical computations, the action-angle variables associated with a time-independent potential, in terms of which we characterize the steady states, and, finally, the observables on which our analysis of the mixing phenomenon

is based.

We consider a collisionless gas consisting of identical particles which are subject to a time-independent external potential Φ_{ext} and to the gravitational potential Φ_{gas} generated by the gas itself. Since all the gas particles have the same mass, it is convenient to normalize the DF such that $f(t,x,p)d^3x\,d^3p$ is the mass of the gas contained in the volume element $d^3x\,d^3p$ of phase space, where from now on p denotes the momentum per unit mass.[2] With these conventions, the Vlasov-Poisson system consists of the Vlasov equation

$$\frac{\partial f}{\partial t} + p\cdot\nabla_x f - (\nabla_x\Phi)\cdot\nabla_p f = 0, \qquad \Phi := \Phi_{ext} + \Phi_{gas}, \tag{1}$$

and the Poisson equation

$$\Delta\Phi_{gas} = 4\pi G\rho, \qquad \rho(t,x) := \int\limits_{\mathbb{R}^3} f(t,x,p)d^3p, \tag{2}$$

where G denotes Newton's constant and ρ is the mass density of the gas. To obtain a unique solution of Eq. (2) we require that $\Phi_{gas}(t,x)\to 0$ for $|x|\to\infty$.

For the external potential we choose Hénon's isochrone potential [1, 52, 53]

$$\Phi_{ext}(x) = -\frac{GM}{b+\sqrt{b^2+r^2}}, \qquad r := |x|, \tag{3}$$

which is generated by a spherical matter distribution of total mass M. The parameter $b\geq 0$ is a length scale which describes the transition from a harmonic potential (when $r\ll b$) to the potential of a point particle of mass M (when $r\gg b$). For $b=0$, Eq. (3) reduces to the Kepler potential $\Phi_{ext} = -GM/r$, which is the external potential considered in Refs. [31, 35]. The reason for this choice is that in this potential the period of the radial motion depends only on the energy of the particle [52, 53], and that the corresponding action-angle variables are known in closed form [1]. When the self-gravity of the gas is neglected, this provides exact solutions against which our numerical methods can be validated.

A. Reduction to spherical symmetry

A spherically symmetric gas configuration is described by a DF which is invariant with respect to simultaneous rotations of x and p. Such a DF depends on (x,p) only through the radius r, the radial component $p_r := x\cdot p/r$ of the momentum and the magnitude $L := |x\wedge p|$ of the angular momentum, that is,

$$f(t,x,p) = \mathcal{F}(t,r,p_r,L). \tag{4}$$

In this case, the density ρ and the potential Φ_{gas} depend only on t and r, and the Vlasov equation (1) reduces to

$$\frac{\partial\mathcal{F}}{\partial t} + p_r\frac{\partial\mathcal{F}}{\partial r} - \frac{\partial V_L}{\partial r}\frac{\partial\mathcal{F}}{\partial p_r} = 0, \qquad V_L(t,r) := \Phi(t,r) + \frac{L^2}{2r^2}, \tag{5}$$

with V_L the effective potential for the radial motion. Since no derivative with respect to L appears in Eq. (5), the gas particles with a given value of L behave like a one-dimensional gas in the potential V_L. However, particles with different values of L interact with each other through the potential Φ_{gas}, which is determined by the spherically reduced Poisson equation

$$\frac{1}{r^2}\frac{\partial}{\partial r}\left(r^2\frac{\partial\Phi_{gas}}{\partial r}\right) = 4\pi G\rho, \qquad \rho(t,r) = \frac{2\pi}{r^2}\int\limits_0^\infty\int\limits_{-\infty}^\infty \mathcal{F}(t,r,p_r,L)\,dp_r\,LdL, \tag{6}$$

see for instance Ref. [17]. The solution of Eq. (6) which is regular at the center and vanishes at infinity is

$$\Phi_{gas}(t,r) = -4\pi G\int\limits_0^\infty \frac{\rho(t,s)}{\max\{r,s\}}s^2ds, \tag{7}$$

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[2] This is equivalent to working in units for which the mass of each gas particle is one, as in Ref. [14].

and its gradient is $\partial\Phi_{gas}/\partial r = Gm(t,r)/r^2$, where $m(t,r) := 4\pi\int_0^r \rho(t,s)s^2 ds$ is the mass of the gas contained in the sphere of radius r at time t.

For the following, it is useful to note that when a spherically symmetric function is integrated over phase space, the volume element can be replaced with

$$d^3x\, d^3p = 8\pi^2 dr\, dp_r\, LdL. \tag{8}$$

In particular, the total mass and the total energy of the gas are given by

$$M_{gas} = 8\pi^2 \int_0^\infty\int_{-\infty}^\infty\int_0^\infty \mathcal{F}(t,r,p_r,L)\, dr\, dp_r\, LdL, \tag{9}$$

$$\mathcal{E}_{gas} = 8\pi^2 \int_0^\infty\int_{-\infty}^\infty\int_0^\infty \mathcal{F}(t,r,p_r,L)\left[\frac{p_r^2}{2} + \frac{L^2}{2r^2} + \Phi_{ext}(r) + \frac{1}{2}\Phi_{gas}(t,r)\right] dr\, dp_r\, LdL, \tag{10}$$

both of which are conserved in time. The conservation of $\mathcal{E}_{gas}$ provides a test for the accuracy of our numerical evolutions, see section V.

B. The reduced model with fixed angular momentum

A further simplification, which has been considered in Refs. [14, 17, 31], is obtained by assuming that all the gas particles have the same magnitude $L_0 > 0$ of the angular momentum.[3] In this case, the DF is of the form

$$\mathcal{F}(t,r,p_r,L) = \mathcal{F}_0(t,r,p_r)\delta(L-L_0), \tag{11}$$

and Eqs. (5,6) yield

$$\frac{\partial\mathcal{F}_0}{\partial t} + p_r\frac{\partial\mathcal{F}_0}{\partial r} - \frac{\partial V_{L_0}}{\partial r}\frac{\partial\mathcal{F}_0}{\partial p_r} = 0, \qquad \rho(t,r) = \frac{2\pi L_0}{r^2}\int_{-\infty}^\infty \mathcal{F}_0(t,r,p_r)\, dp_r, \tag{12}$$

where the potential Φ_{gas} is still given by Eq. (7). Likewise, Eqs. (9,10) hold with $\mathcal{F}$ replaced by $\mathcal{F}_0$, L by L_0, and the integral over L by the factor L_0. Equations (12,7) constitute a Vlasov-Poisson system on a two-dimensional phase space with coordinates (r, p_r). It describes a gas of concentric spherical shells which move in the effective potential V_{L_0}. The centrifugal term in this potential prevents the shells from reaching the center, and the gas configurations considered in these notes are confined to a region of the form $0 < r_{in} \le r \le r_{out}$. In spite of its simplicity, the reduced model retains the two ingredients which are relevant for these notes, namely the dependency of the frequency of the radial motion on the orbit, which is responsible for the mixing, and the self-gravity of the gas.

C. Dimensionless variables

In our numerical computations we use dimensionless variables, which are obtained by measuring masses in units of M, lengths in units of b and times in units of the dynamical time

$$t_{dyn} := \sqrt{\frac{b^3}{GM}}. \tag{13}$$

Accordingly, p_r is measured in units of $\sqrt{GM/b}$, L in units of $\sqrt{GMb}$, the potentials and the energy per unit mass in units of GM/b, and $\mathcal{F}$ in units of $M/(GMb)^{3/2}$. In terms of these variables, all the equations of this section retain their form with $G = M = b = 1$; in particular, the isochrone potential is $\Phi_{ext}(r) = -1/(1+\sqrt{1+r^2})$. The resulting system contains no free parameters, and the strength of the self-gravity is controlled by the ratio M_{gas}/M between

[3] Note that in Ref. [31] the symbol L refers to the square of the magnitude of the angular momentum.

the mass of the gas and the mass which generates the external potential. In the limit $M_{gas}/M \to 0$ the potential Φ_{gas} can be neglected, and one recovers the model analyzed in Ref. [14].

In the Kepler case $b = 0$ the external potential does not provide a length scale. However, in this case the system (5,6) is invariant with respect to the rescaling

$$(t, r, p_r, L, \mathcal{F}, \Phi) \mapsto \left(\alpha^{3/2} t, \alpha r, \alpha^{-1/2} p_r, \alpha^{1/2} L, \alpha^{-3/2} \mathcal{F}, \alpha^{-1} \Phi\right), \qquad \alpha > 0, \tag{14}$$

which leaves M_{gas} unchanged. Therefore, b can be replaced with an arbitrary length in the definitions above, such that $\Phi_{ext}(r) = -1/r$, and in the reduced model this freedom can be used to fix the value of L_0.

To give an idea of the physical time scale, we note that $M = 10^5 M_\odot$ and $b = 1$ pc, which are of the order of magnitude of the mass and of the core radius of a globular cluster [1], yield $t_{dyn} \approx 4.7 \times 10^4$ yr. As will be seen below, $2\pi t_{dyn}$ is a lower bound for the periods of the radial motion in the isochrone potential.

D. Action-angle variables and steady states

Suppose that the total potential Φ is independent of time. In our applications, Φ is the external potential (3), the total potential Φ_{eq} of a self-consistent steady state (defined below), or the potential of an evolving gas at a fixed instant of time, see section II E. In all these cases $\Phi'(r) = G m_{tot}(r)/r^2$, where $m_{tot}(r)$ denotes the total mass contained in the sphere of radius r, including the one which generates the external potential. The energy per unit mass

$$E := \frac{p_r^2}{2} + V_L(r) \tag{15}$$

is conserved along the trajectories of the gas particles, and following Refs. [1, 14] we introduce action-angle variables for their radial motion. Since the function $r \mapsto r m_{tot}(r)$ increases strictly monotonically from 0 to ∞, for each $L > 0$ the effective potential V_L has a unique critical point $r = r_{min}(L)$. It is determined by the equation $r^3 \Phi'(r) = L^2$, it is a non-degenerate minimum of V_L, and it corresponds to a circular orbit with energy $E_{min}(L) := V_L(r_{min}(L))$. Since $V_L(r) \to \infty$ for $r \to 0$ and $V_L(r) \to 0$ for $r \to \infty$, the orbits with energies in the range $E_{min}(L) < E < 0$ are bound, and they oscillate between the turning points $r_1(E, L) < r_2(E, L)$, which are the two solutions of the equation $V_L(r) = E$.

As in Ref. [14] we define the area function

$$A(E, L) := \oint p_r dr = 2 \int_{r_1(E,L)}^{r_2(E,L)} \sqrt{2(E - V_L(r))}\, dr, \tag{16}$$

which represents the area enclosed by the orbit in the (r, p_r)-plane, and whose partial derivative with respect to E yields the period of the radial motion,

$$T(E, L) := \frac{\partial A}{\partial E}(E, L) = 2 \int_{r_1(E,L)}^{r_2(E,L)} \frac{dr}{\sqrt{2(E - V_L(r))}}. \tag{17}$$

The action and angle variables associated with the radial motion are

$$J := \frac{A(E, L)}{2\pi}, \qquad Q := \frac{2\pi}{T(E, L)} \int_{\gamma_r} \frac{dr}{p_r}, \tag{18}$$

where γ_r denotes the segment of the orbit in the (r, p_r)-plane which connects, in the clockwise direction, the inner turning point $(r_1(E, L), 0)$ with the given point (r, p_r). Hence, Q/Ω with $\Omega := 2\pi/T(E, L)$ is the time which has elapsed since the last passage of the particle through its pericenter; in particular, $Q = 0$ at the pericenter, $Q = \pi$ at the apocenter, and $0 < Q < \pi$ when $p_r > 0$. For each fixed value of L, the map $(r, p_r) \mapsto (Q, J)$ is symplectic, that is $dr \wedge dp_r = dQ \wedge dJ$, and it provides smooth coordinates on the set of bound orbits which are not circular. Since $T > 0$, the first relation in Eq. (18) can be inverted for each L, which yields the energy as a function $E(J, L)$ of the action variable and the angular momentum, and the frequency of the radial motion is

$$\Omega(J, L) = \frac{\partial E}{\partial J}(J, L) = \frac{2\pi}{T(E, L)}. \tag{19}$$

[(Erik) We denote the orbital frequency by Ω (it is called ω in Ref. [14]) and reserve ω for the frequency of the collective oscillation.] In terms of the variables (Q, J, L), the volume element (8) is $8\pi^2 dQ\, dJ\, LdL$, and the Vlasov equation (5) for a time-independent potential assumes the simple form

$$\frac{\partial \mathcal{F}}{\partial t} + \Omega(J, L)\frac{\partial \mathcal{F}}{\partial Q} = 0. \tag{20}$$

In particular, the time-independent solutions of Eq. (20) are the DFs which do not depend on the angle variable. Since $E(J, L)$ is a strictly increasing function of J, they are of the form

$$\mathcal{F}(r, p_r, L) = \psi(E, L), \tag{21}$$

with a nonnegative function ψ. Such a DF is a solution of the Vlasov-Poisson system if the potential Φ_{gas}, which enters the definition of E through V_L, is the one generated by the density of the DF through Eqs. (6,7). This is a nonlinear problem for Φ_{gas}, whose numerical solution is described in section IV B. A solution $(\mathcal{F}_{eq}, \Phi_{eq})$ of this problem, with Φ_{eq} the total potential and ρ_{eq} the corresponding density, will be referred to as a self-consistent steady state.

For the external potential (3) the area function can be computed explicitly, see Refs. [1, 14]. In terms of the dimensionless variables introduced in the previous subsection, one finds

$$A(E, L) = 2\pi\left[\frac{1}{\sqrt{-2E}} - c(L)\right], \qquad E_{min}(L) = -\frac{1}{2c(L)^2}, \tag{22}$$

with $c(L) := \left(L + \sqrt{L^2 + 4}\right)/2$ for the isochrone potential and $c(L) := L$ for the Kepler potential. This yields

$$E(J, L) = -\frac{1}{2\left[J + c(L)\right]^2}, \qquad \Omega(J, L) = \frac{1}{\left[J + c(L)\right]^3} = (-2E)^{3/2}. \tag{23}$$

Therefore, the frequency depends only on the energy, it decreases monotonically when J or L increase, and for each fixed L its maximum is attained in the limit $J \to 0$ of circular orbits. For the isochrone potential, $c(L) > 1$ implies that $\Omega < 1$, such that the periods of the radial motion are larger than 2π. An explicit expression for the angle variable Q in terms of (r, p_r, L) can be found in section 3.5.2 of Ref. [1]; its inversion requires the solution of an equation of Kepler's type. In contrast to this, when the self-gravity of the gas is taken into account the potential is only known numerically, and in this case we compute the action-angle variables by numerical quadrature of Eqs. (16,17,18), see Appendix A.

E. Observables

As in Ref. [14], we describe the macroscopic state of the gas through observables of the form

$$N_\varphi(t) := \int f(t, x, p)\varphi(x, p)\, d^3x\, d^3p, \tag{24}$$

where φ is a given test function on phase space. In these notes we consider two classes of such observables.

The first one refers to the action-angle variables (Q, J) belonging to a time-independent potential. For initial data close to a self-consistent steady state, this is the total potential Φ_{eq} of that state, which reduces to Φ_{ext} when the self-gravity of the gas is neglected. For initial data far from equilibrium there is no such state, and we use the action-angle variables of the instantaneous potential: at each time t, (Q, J) are the action-angle variables which belong to the potential $r \mapsto \Phi(t, r)$, regarded as a time-independent potential. For a spherically symmetric configuration and a test function of the form $\varphi = a(Q)B(J, L)$, with real-valued functions a and B, Eqs. (8,24) and the expansion of a in a Fourier series yield

$$N_\varphi(t) = \sum_{n\in\mathbb{Z}} \hat{a}_n^*\, h_n(t), \qquad \hat{a}_n := \frac{1}{2\pi}\int_0^{2\pi} a(Q)e^{-inQ}dQ, \tag{25}$$

with the functions

$$h_n(t) := 8\pi^2 \int_0^\infty\int_0^\infty\int_0^{2\pi} \mathcal{F}(t, Q, J, L)B(J, L)e^{-inQ}\, dQ\, dJ\, LdL, \qquad n \in \mathbb{Z}. \tag{26}$$

In the reduced model, B depends only on J and Eq. (26) holds with $\mathcal{F}$ replaced by $\mathcal{F}_0$ and the integral over L by the factor L_0. Since $\mathcal{F}$ is real, $h_{-n} = h_n^*$, such that it is sufficient to consider $n \geq 0$. For a steady state h_n vanishes for all $n \neq 0$, whereas h_0 is equal to the mass of the gas weighted with the function B. Therefore, the functions $h_n(t)$ with $n \neq 0$ measure the departure of the gas configuration from a steady state, and their decay in time is the signature of the relaxation process we are interested in. We stress that this property relies on the use of the action-angle variables which belong to the total potential. If those of the external potential are used instead, a self-consistent steady state gives rise to nonvanishing, time-independent values of h_n with $n \neq 0$, as we show in section V. When the action-angle variables of the instantaneous potential are used, the test function depends on the DF through $\Phi(t, \cdot)$, such that h_n is no longer a linear functional of the DF. On the other hand, h_n with $n \neq 0$ then vanishes for any steady state, including a steady state which the gas may approach at late times and which is not known in advance.

The second class of observables consists of the perturbation of the gravitational potential,

$$\delta\Phi(t, r) := \Phi(t, r) - \Phi_{eq}(r) = -4\pi G \int_0^\infty \frac{\rho(t, s) - \rho_{eq}(s)}{\max\{r, s\}} s^2 ds, \tag{27}$$

where we have used Eq. (7). For each fixed r this is the difference between the values of the observable (24) with test function $\varphi(x, p) = -G/\max\{r, |x|\}$ which belong to the DF of the gas and to the one of the steady state. In contrast to the functions $h_n(t)$, the quantity $\delta\Phi$ does not refer to any particular coordinates on phase space. For initial data far from equilibrium we replace Φ_{eq} in Eq. (27) with the average of Φ over a late time interval, which is specified in section VII. The results in Refs. [31, 32] refer to the decay of the force field $-\partial\delta\Phi/\partial r$ which is generated by it, or of the time derivative of this field. To measure its magnitude we use the root mean square

$$\|\delta\Phi(t)\| := \left(\frac{1}{r_b - r_a} \int_{r_a}^{r_b} \delta\Phi(t, r)^2 dr \right)^{1/2} \tag{28}$$

over an interval $[r_a, r_b]$ which covers the region occupied by the gas.

III. MIXING AND LINEAR RESPONSE

In this section we first review the mixing phenomenon in the absence of self-gravity, for an arbitrary initial DF, and discuss the rate at which the observables introduced in the previous section decay. Next, we linearize the Vlasov-Poisson system about a self-consistent steady state, which describes initial data close to that steady state, and state the criterion for the existence of oscillating modes, which determines whether or not the mixing is completed in the linear regime. We end the section with the role played by the edge of the support of the steady state. The derivations in this section are formal; for rigorous statements we refer the reader to Refs. [14, 16] regarding the mixing property and to Refs. [26, 27, 30, 31] regarding the linearized system.

A. Mixing in the absence of self-gravity

When the self-gravity of the gas is neglected, $\Phi = \Phi_{ext}$ is independent of time, and the solution of Eq. (20) is

$$\mathcal{F}(t, Q, J, L) = F(Q - \Omega(J, L)t, J, L), \tag{29}$$

where the function F, which is 2π-periodic in its first argument, describes the initial DF. No assumption on F is needed here: it need not be close to a steady state. Expanding F in a Fourier series with respect to the angle variable,

$$F(Q, J, L) = \sum_{n \in \mathbb{Z}} \hat{F}_n(J, L) e^{inQ}, \qquad \hat{F}_n(J, L) := \frac{1}{2\pi} \int_0^{2\pi} F(Q, J, L) e^{-inQ} dQ, \tag{30}$$

and introducing Eq. (29) into Eq. (26) one obtains

$$h_n(t) = 16\pi^3 \int_0^\infty \int_0^\infty \hat{F}_n(J, L) B(J, L) e^{-in\Omega(J,L)t} \, dJ \, L dL. \tag{31}$$

Hence, h_0 is constant in time, whereas for $n \neq 0$ the function $h_n(t)$ is a superposition of oscillations whose frequencies $n\Omega(J, L)$ vary from one orbit to the other. Provided that the gradient of Ω with respect to (J, L) is different from zero for almost all (J, L) (in the reduced model, provided that $\partial\Omega/\partial J \neq 0$ for almost all J), it follows from the Riemann-Lebesgue lemma and its generalizations (cf. Refs. [14, 16]) that $h_n(t) \to 0$ for $t \to \infty$ and all $n \neq 0$. As a consequence, the observables (25) converge to the ones which belong to the angle-averaged DF $\hat{F}_0(J, L)$, which is a steady state according to the remark below Eq. (20). This is the phase space mixing property, and $\hat{F}_0$ is the final state of the gas. For the isochrone and the Kepler potentials, Eq. (23) gives $\partial\Omega/\partial J = -3[J + c(L)]^{-4} < 0$, such that the non-degeneracy condition is satisfied.

The rate at which $h_n(t)$ decays is not universal. It depends on the regularity of the function $\hat{F}_n B$ and on its behavior at the boundary of its support. If this function is smooth and vanishes together with all its derivatives at the boundary, repeated integration by parts in Eq. (31) shows that $h_n(t)$ decays faster than any inverse power of t. A different situation, which is the relevant one for most of the initial data considered in these notes, arises in the reduced model when the DF has its support in an interval $0 \leq J \leq J_{max}$ and vanishes like a power at the outer edge, such that

$$\hat{F}_n(J)B(J) = (J_{max} - J)^k g(J) \tag{32}$$

with an exponent $k > 0$ and a smooth function g satisfying $g(J_{max}) \neq 0$. Suppose that $\Omega'(J) < 0$ for all $0 < J \leq J_{max}$, as is the case in all the examples of these notes, and that g vanishes sufficiently fast for $J \to 0$. Then, the behavior of the integral in Eq. (31) for large times is determined by the endpoint $J = J_{max}$ (see, for instance, chapter 6 in Ref. [54]), and for $n \geq 1$ one finds

$$h_n(t) \simeq 16\pi^3 L_0\, \Gamma(k+1)\, g(J_{max}) \frac{e^{-in\Omega_{min}t}}{[in|\Omega'(J_{max})|t]^{k+1}}, \qquad t \to \infty, \tag{33}$$

where $\Omega_{min} := \Omega(J_{max})$ is the frequency of the outermost orbit and Γ denotes the Gamma function. Therefore, an edge at which the DF vanishes with the exponent k gives rise to an algebraic tail which is proportional to $t^{-(k+1)}$ and which oscillates with the frequency $n\Omega_{min}$: the less regular the DF is at the edge, the slower is the decay. The general rule behind this result is that a singularity of the form $(\omega - \omega_0)^a$ of the Laplace transform of an observable (with a logarithmic factor when a is an integer) gives rise to a tail which is proportional to $t^{-(a+1)}$ and oscillates with the frequency ω_0; the singularities which occur for systems with one and with two action variables have been classified in Refs. [51, 55]. Decay estimates for the mixing in the potential of a steady state whose DF vanishes like a power at the edge of its support have been proven in Ref. [23].

When the angular momenta of the gas particles are distributed, the integration over L in Eq. (31) provides an additional source of decay, since Ω depends on L as well. For the isochrone and the Kepler potentials, Eq. (23) yields $\partial\Omega/\partial L = c'(L)\,\partial\Omega/\partial J$ with $1/2 < c'(L) \leq 1$, such that a dispersion in L is almost as efficient for the mixing as a dispersion of the same magnitude in J. In these notes we restrict ourselves to the reduced model.

B. Linearization about a self-consistent steady state

When the self-gravity of the gas is taken into account, the potential depends on time, and there is no longer a fixed set of action-angle variables in which the motion of the gas particles is trivial. If the initial DF is close to a self-consistent steady state $(\mathcal{F}_{eq}, \Phi_{eq})$, it is convenient to use the action-angle variables (Q, J) which belong to the potential Φ_{eq}, such that $\mathcal{F}_{eq}$ is a function of (J, L) only. We write

$$\mathcal{F} = \mathcal{F}_{eq} + \delta\mathcal{F}, \qquad \Phi = \Phi_{eq} + \delta\Phi, \tag{34}$$

with $\delta\Phi$ the perturbation of the potential defined in Eq. (27). The motion of the gas particles is determined by the Hamiltonian $H = E(J, L) + \delta\Phi(t, r(Q, J, L))$, where $r(Q, J, L)$ denotes the radius as a function of the action-angle variables. Since for each fixed L the variables (Q, J) are canonical, the Vlasov equation (5) is equivalent to

$$\frac{\partial\delta\mathcal{F}}{\partial t} + \Omega\frac{\partial\delta\mathcal{F}}{\partial Q} - \frac{\partial\mathcal{F}_{eq}}{\partial J}\frac{\partial\delta\Phi}{\partial Q} = \frac{\partial\delta\Phi}{\partial Q}\frac{\partial\delta\mathcal{F}}{\partial J} - \frac{\partial\delta\Phi}{\partial J}\frac{\partial\delta\mathcal{F}}{\partial Q}, \tag{35}$$

where no approximation has been made so far. The right-hand side is quadratic in the deviation from the steady state, and neglecting it yields the linearized equation

$$\frac{\partial\delta\mathcal{F}}{\partial t} + \Omega(J, L)\frac{\partial\delta\mathcal{F}}{\partial Q} - \frac{\partial\mathcal{F}_{eq}}{\partial J}(J, L)\frac{\partial\delta\Phi}{\partial Q} = 0. \tag{36}$$

The potential is determined by Eqs. (6,7,8), which give

$$\delta\Phi(t,r) = -8\pi^2 G \int_0^\infty \int_0^\infty \int_0^{2\pi} \frac{\delta\mathcal{F}(t,Q',J',L')}{\max\{r, r(Q',J',L')\}}\, dQ'\, dJ'\, L' dL'. \tag{37}$$

The second term on the left-hand side of Eq. (36) describes the transport along the orbits of the steady state discussed in the previous subsection. The third term describes the reaction of the steady state to the perturbation of the potential: the latter changes the action variables of the gas particles at the rate $\dot{J} = -\partial\delta\Phi/\partial Q$, and this displaces mass along the gradient of $\mathcal{F}_{eq}$. Through this term, which is absent when self-gravity is neglected, all the orbits are coupled to each other. When the initial data are far from any steady state, as in section VII, the right-hand side of Eq. (35) is not small, and the evolution has to be followed with the nonlinear equations.

Since the support of $\mathcal{F}_{eq}$ is bounded, the frequencies $\Omega(J,L)$ of the orbits which are populated by the steady state fill an interval $[\Omega_{min}, \Omega_{max}]$ with $0 < \Omega_{min} < \Omega_{max}$, which will be referred to as the frequency band in what follows. Without the third term in Eq. (36), the n-th Fourier coefficient of $\delta\mathcal{F}$ with respect to Q oscillates with the frequencies $n\Omega(J,L)$, which lie in the interval $n[\Omega_{min}, \Omega_{max}]$. These intervals constitute the essential spectrum of the linearized system, which is not altered by the self-gravity of the perturbation [26, 31].

C. Oscillating modes and the gain of the self-consistent loop

In the linear regime, the mixing of a deviation from the steady state is not completed if the linearized system possesses solutions of the form $\delta\mathcal{F} = a(Q,J,L)e^{-i\omega t} + c.c.$ with a real frequency $\omega > 0$, where $c.c.$ denotes the complex conjugate of the preceding term. Such a solution describes a perturbation which oscillates without being damped, and it will be called an oscillating mode in what follows. Expanding $a(Q,J,L) = \sum_n a_n(J,L)e^{inQ}$, and the potential along the orbits in the same way, with coefficients $\phi_n(J,L) = \phi_{-n}(J,L)$, Eq. (36) yields

$$[n\Omega(J,L) - \omega]\, a_n(J,L) = n\frac{\partial\mathcal{F}_{eq}}{\partial J}(J,L)\, \phi_n(J,L), \qquad n \in \mathbb{Z}. \tag{38}$$

We assume that ω lies below the frequency band, $0 < \omega < \Omega_{min}$, such that none of the gas particles is in resonance with the perturbation. Since the radius is the same at the two points of an orbit with angle variables Q and $2\pi - Q$, the potential generated by the perturbation depends only on the combinations $b_n := a_n + a_{-n}$ with $n \geq 1$, for which Eq. (38) gives

$$b_n(J,L) = R_n(J,L;\omega)\, \phi_n(J,L), \qquad R_n(J,L;\omega) := \frac{2n^2\Omega(J,L)}{n^2\Omega(J,L)^2 - \omega^2}\frac{\partial\mathcal{F}_{eq}}{\partial J}(J,L), \tag{39}$$

while Eq. (37) yields

$$\phi_n(J,L) = \sum_{n'=1}^{\infty} \int_0^\infty \int_0^\infty M_{nn'}(J,L;J',L')\, b_{n'}(J',L')\, dJ'\, L' dL', \tag{40}$$

with the kernel

$$M_{nn'}(J,L;J',L') := -4\pi G \int_0^{2\pi} \int_0^{2\pi} \frac{\cos(nQ)\cos(n'Q')}{\max\{r(Q,J,L), r(Q',J',L')\}}\, dQ\, dQ'. \tag{41}$$

Hence, an oscillating mode with frequency ω exists if and only if one is an eigenvalue of the operator $MR(\omega)$, which maps a perturbation of the potential with frequency ω to the potential generated by the response of the gas. In the reduced model, these equations hold with $L = L' = L_0$, the function $\mathcal{F}_{eq}$ replaced by $\mathcal{F}_{0,eq}(J)$, and the integral over L' by the factor L_0.

The operator M is symmetric and negative with respect to the natural inner product on the functions $b = (b_n)_{n\geq 1}$, since the gravitational interaction is attractive. If, in addition, $\partial\mathcal{F}_{eq}/\partial J < 0$ in the interior of the support of the steady state, which is Antonov's stability condition [1, 31] and holds for all the steady states considered in these notes, then $R_n(J,L;\omega) < 0$ for $0 \leq \omega < \Omega_{min}$, and $MR(\omega)$ has the same nonvanishing eigenvalues as the symmetric and nonnegative operator

$$S(\omega) := |R(\omega)|^{1/2}(-M)|R(\omega)|^{1/2}. \tag{42}$$

We denote by $\lambda(\omega)$ the largest eigenvalue of $S(\omega)$ and call it the gain of the self-consistent loop. Since $|R_n(J, L; \omega)|$ increases with ω, the function $\lambda(\omega)$ is nondecreasing on the interval $[0, \Omega_{min})$, and we define its value at the edge of the band by

$$\lambda_{edge} := \lim_{\omega \to \Omega_{min}} \lambda(\omega) \in (0, \infty]. \tag{43}$$

The same construction applies to solutions which grow exponentially in time, for which ω^2 is replaced with a negative number in Eq. (39), and it shows that such solutions are absent when $\lambda(0) < 1$; for all the steady states of section VI we find $\lambda(0) < 0.6$. For a steady state with $\lambda(0) < 1$ one arrives at the following criterion. If $\lambda_{edge} > 1$, there exists an oscillating mode whose frequency ω_d lies below the frequency band and is determined by

$$\lambda(\omega_d) = 1, \qquad 0 < \omega_d < \Omega_{min}. \tag{44}$$

If $\lambda_{edge} < 1$, there are no oscillating modes with frequencies below the band. In the first case the mixing of a deviation from the steady state is not completed, since the part of the deviation which is carried by the mode persists. This criterion is a formulation in terms of action-angle variables of the Birman-Schwinger principle of Refs. [26–29, 31]; $\lambda(\omega)$ coincides with the norm of the operator of Mathur, and λ_{edge} corresponds to the quantity which is denoted by M in Ref. [26] and by μ_* in Ref. [28]. In section IV we describe a numerical method for the computation of $\lambda(\omega)$, which allows us to evaluate the criterion for steady states of arbitrary mass. Note that the criterion presupposes the existence of the gap $0 < \omega < \Omega_{min}$, which requires that the support of the steady state does not reach orbits of arbitrarily small frequency; all the steady states considered in these notes have compact support.

When $\lambda_{edge} < 1$, the deviation from the steady state decays in all the examples of section VI. When the self-gravity of the gas is not too weak, the response to an initial deviation is dominated at intermediate times by a damped oscillation proportional to $e^{-i\omega t - \gamma t}$ with a frequency ω inside the band and a damping rate $\gamma > 0$, which is followed at late times by an algebraic tail. The damped oscillation is the analogue of Landau damping in a plasma [24]. It does not correspond to an eigenvalue of the linearized system; instead, $\omega - i\gamma$ is a zero of the analytic continuation of the function $\omega \mapsto \det[1 - MR(\omega)]$ from the upper to the lower half of the complex plane, see for instance Refs. [1, 56, 57]. We do not compute this continuation in these notes; instead, we determine ω and γ by fitting the time series which are obtained from the numerical solution of the linearized equations and from the PIC code, as described in section IV.

D. Behavior at the edge of the frequency band

The edge of the support of the steady state controls both the tail of the mixing, Eq. (33), and the value of λ_{edge}. Consider the reduced model and a steady state whose DF has a cutoff at the energy E_0 and behaves like

$$\mathcal{F}_{0,eq} \simeq \alpha (E_0 - E)^k, \qquad E \to E_0, \tag{45}$$

with constants $\alpha > 0$ and $k > 1/2$, as is the case for the polytropes considered in Ref. [31]. Let J_{max} be the action variable of the orbit with energy E_0, such that $\Omega_{min} = \Omega(J_{max})$. Then $\partial \mathcal{F}_{0,eq}/\partial J$ is proportional to $(J_{max} - J)^{k-1}$ near the edge, and Eq. (39) with $n = 1$ and $\omega = \Omega_{min}$ yields

$$|R_1(J; \Omega_{min})| \simeq \frac{1}{\Omega(J) - \Omega_{min}} \left| \frac{\partial \mathcal{F}_{0,eq}}{\partial J}(J) \right| \propto (J_{max} - J)^{k-2}, \qquad J \to J_{max}. \tag{46}$$

This function is integrable at $J = J_{max}$ if and only if $k > 1$. If $k > 1$, λ_{edge} is finite and proportional to M_{gas}/M as long as this ratio is small, such that an oscillating mode only appears above a threshold mass; this threshold is computed in section VI. If $1/2 < k \leq 1$, $\lambda(\omega)$ diverges for $\omega \to \Omega_{min}$, such that $\lambda_{edge} = \infty$ and an oscillating mode exists for any mass of the gas, as proven in Ref. [31] for small masses. However, since the factor which multiplies the divergent term is proportional to M_{gas}/M, Eq. (44) implies that, to leading order in the limit $M_{gas}/M \to 0$,

$$\Omega_{min} - \omega_d \propto \left(\frac{M_{gas}}{M} \right)^{\frac{1}{1-k}} \quad (k < 1), \qquad \Omega_{min} - \omega_d \propto \exp\left(-C \frac{M}{M_{gas}} \right) \quad (k = 1), \tag{47}$$

with a positive constant C. Since a time of the order of $(\Omega_{min} - \omega_d)^{-1}$ is required to distinguish the mode from the response of the orbits close to the edge, which oscillates with a frequency close to Ω_{min} and decays algebraically, for a gas of small mass the mode cannot be observed in a time evolution of practical duration, and the mixing appears to be completed.

IV. NUMERICAL METHODS

In this section we describe the numerical methods on which our study is based. We start with the PIC method used to solve the nonlinear Vlasov-Poisson system and with the construction of the self-consistent steady states. Next, we explain how the initial data, both far from equilibrium and close to a steady state, are represented by the computational particles, and how in the second case the response of the gas to the perturbation is extracted from the simulations. Finally, we describe the numerical solution of the linearized equations, the computation of the gain $\lambda(\omega)$ of the self-consistent loop, and the determination of frequencies and damping rates from a time series.

A. Particle-in-cell method

In the PIC method [45, 46] the DF is represented by a finite number N of computational particles, each of which stands for a large number of gas particles. In spherical symmetry a computational particle is a spherical shell, which is characterized by its radius $r_j(t)$, its radial momentum $p_j(t)$, its angular momentum L_j and its mass m_j, $j = 1, 2, \ldots, N$, the last two quantities being constant in time. This corresponds to the approximation

$$\mathcal{F}(t, r, p_r, L) \approx \sum_{j=1}^{N} \frac{m_j}{8\pi^2 L_j}\, \delta(r - r_j(t))\, \delta(p_r - p_j(t))\, \delta(L - L_j) \tag{48}$$

of the DF, where the factor $8\pi^2 L_j$ is due to the volume element (8). Introducing Eq. (48) into the Vlasov equation (5) yields the equations of motion

$$\frac{dr_j}{dt} = p_j, \qquad \frac{dp_j}{dt} = \frac{L_j^2}{r_j^3} - \frac{\partial \Phi_{ext}}{\partial r}(r_j) - \frac{\partial \Phi_{gas}}{\partial r}(t, r_j) \tag{49}$$

for the computational particles; in the reduced model $L_j = L_0$ for all j. The first two terms on the right-hand side of the second equation are evaluated analytically, whereas the last one requires the solution of the Poisson equation (6). To this purpose we introduce the uniform grid $R_i := (i - 1/2)\Delta r$, $i = 1, 2, \ldots, N_r$, which is staggered with respect to the center, and proceed in four steps.

(i) The mass of each computational particle is assigned to its two neighboring grid points with the linear weights of the cloud-in-cell scheme [45, 46], such that the grid point R_i receives the mass

$$\mu_i := \sum_{j=1}^{N} m_j W\left(\frac{R_i - r_j}{\Delta r}\right), \qquad W(y) := \max\{1 - |y|, 0\}. \tag{50}$$

(ii) The density is approximated by the continuous, piecewise linear function $\rho(t, r)$ which assumes the values

$$\rho_i := \frac{\mu_i}{4\pi \Delta r\, (R_i^2 + \Delta r^2/6)} \tag{51}$$

at the grid points and which is constant for $0 \le r \le R_1$. The denominator in Eq. (51) is the integral of the function $4\pi r^2 W((r - R_i)/\Delta r)$ over r, which guarantees that the mass of the reconstructed density coincides with the total mass of the computational particles.

(iii) The mass function $m(t, r)$ and the potential are obtained by integrating the equations $\partial m/\partial r = 4\pi r^2 \rho$ and $\partial \Phi_{gas}/\partial r = Gm/r^2$ outwards from the center. Since ρ is linear on each interval $[R_{i-1}, R_i]$, both integrals are computed in closed form, such that no further discretization error is introduced. The additive constant in Φ_{gas} is fixed by requiring that $\Phi_{gas} + r\, \partial \Phi_{gas}/\partial r = 0$ at the outermost grid point, which matches the potential to the exterior solution $-GM_{gas}/r$.

(iv) The potential and its gradient are interpolated from the grid points to the positions of the computational particles with the same weights as in Eq. (50).

For particles which lie within the distance Δr from the center, the weights in steps (i) and (iv) are complemented by those of their mirror images at $-r_j$, in accordance with the regularity of the density at the center, and beyond the outermost grid point the exterior solution is used. Neither of these situations plays a role for the configurations considered in these notes, in which the gas occupies a shell that lies well inside the grid; the computational particles

which leave the grid for the widest initial data of section VII carry less than 10^{-11} of the mass of the gas. For the Kepler potential the code uses the potential of a homogeneous sphere of mass M and unit radius, which coincides with the Kepler potential for $r \geq 1$. In all the simulations of these notes with this potential the gas remains in the region $r > 1.7$, such that the two potentials are equivalent.

Note that the potential which acts on a computational particle includes the contribution of the particle itself. In PIC schemes based on Cartesian coordinates the use of the same weights for the assignment and for the interpolation guarantees that a particle exerts no force on itself [45, 46]. In spherical symmetry this is not the case, and it should not be: a thin shell of mass m and radius r is subject to the mean value of the gravitational fields on its two sides, that is, to the force $-Gm/(2r^2)$ per unit mass. This is the self-gravity of the shell; the scheme reproduces it up to a correction of the order $\Delta r/r$, and we do not subtract it. In fact, we have verified that subtracting it reduces the order of convergence with respect to $1/N$ from two to one in the collapse of a cold homogeneous sphere, for which the exact solution is known.

Equations (49) are integrated in time with the fourth-order symplectic method of Yoshida [58], which is a composition of three steps of the second-order leapfrog method with step sizes $w_1\Delta t$, $w_0\Delta t$ and $w_1\Delta t$, where $w_1 := 1/(2-2^{1/3})$ and $w_0 := 1 - 2w_1$. A leapfrog step with step size h consists of the update $p_j \mapsto p_j + hf_j/2$ of the momenta, with f_j the right-hand side of the second equation in (49), followed by the update $r_j \mapsto r_j + hp_j$ of the radii, the computation of the potential for the new radii, and a second update of the momenta with the new values of f_j. Hence, the Poisson equation is solved three times per time step. We use a symplectic method because, for the motion in a given potential, it does not introduce a secular drift of the energy and of the action variables of the particles, which would otherwise limit the accuracy of the observables at late times. For the same reason we keep the time step Δt fixed.

At selected times we compute the total energy (10),

$$\mathcal{E}_{gas} \approx \sum_{j=1}^{N} m_j \left[\frac{p_j^2}{2} + \frac{L_j^2}{2r_j^2} + \Phi_{ext}(r_j) + \frac{1}{2}\Phi_{gas}(t, r_j) \right], \tag{52}$$

the potential Φ_{gas} on the grid, from which $\delta\Phi$ is obtained according to Eq. (27), and the functions

$$h_n(t) \approx \sum_{j=1}^{N} m_j B(J_j, L_j) e^{-inQ_j}, \tag{53}$$

which follow from Eqs. (26,48). Here (Q_j, J_j) denote the action-angle variables of the particle j at the time t with respect to the potential specified in section II E. They are computed from (r_j, p_j) with the method described in Appendix A; for the instantaneous potential, Φ_{gas} is the potential computed by the PIC code on its grid at the time t. For the initial data close to a steady state we choose the weight function

$$B(J) = J^2 \exp\left[-\frac{(J - J_1)^2}{\sigma_1^2} \right], \tag{54}$$

with parameters J_1 and σ_1 which are specified below. The factor J^2 makes the function $B(J)e^{-inQ}$ continuous at the circular orbit for $|n| \leq 4$. For the initial data far from equilibrium of section VII we choose $B = 1$, such that h_n/h_0 is the average of e^{-inQ} over the mass of the gas.

The simulations of these notes use an implementation of this method for the reduced model. Unless otherwise stated, they use $\Delta r = 0.1$, a grid which extends to $r = 20$, a time step $\Delta t = 0.05$ or 0.1, and $N = 10^4$ computational particles. [(Erik) To be decided: whether the codes and the scripts which reproduce the results are made public, in which case the repository should be cited here.]

B. Self-consistent steady states

As explained in section II D, the construction of a self-consistent steady state requires the solution of a nonlinear problem, because the DF is a prescribed function of integrals of motion which themselves depend on the potential generated by the gas. In the reduced model we consider the following two families of steady states. The first one consists of the polytropes

$$\mathcal{F}_{0,eq} = \alpha(E_0 - E)^k \quad \text{for } E < E_0, \tag{55}$$

and $\mathcal{F}_{0,eq} = 0$ for $E \geq E_0$, with an exponent $k > 1/2$, which are the steady states considered in Ref. [31]. The second family consists of the lowered Maxwellians

$$\mathcal{F}_{0,eq} = \alpha\, E_\gamma\left(g, \frac{E_0 - E}{\vartheta}\right) \quad \text{for } E < E_0, \qquad E_\gamma(g, x) := \sum_{l=0}^{\infty} \frac{x^{g+l}}{\Gamma(g + l + 1)}, \tag{56}$$

and $\mathcal{F}_{0,eq} = 0$ for $E \geq E_0$, see Ref. [59]. They include the King model [60] for $g = 1$, in which case $E_\gamma(1, x) = e^x - 1$, and the Wilson model [61] for $g = 2$, in which case $E_\gamma(2, x) = e^x - 1 - x$. Far from the cutoff the DF (56) is proportional to $e^{-E/\vartheta}$, whereas for $E \to E_0$ it vanishes like $(E_0 - E)^g$, such that g plays the role of the exponent k in Eq. (45). The parameter ϑ is specified through the dimensionless depth $W_0 := [E_0 - E_{min}(L_0)]/\vartheta$ of the potential well. In both families we prescribe the action variable J_{max} of the outermost orbit instead of the cutoff energy, which is then given by $E_0 = E(J_{max}, L_0)$, and we determine the constant α from the mass of the gas. Since the transformation to action-angle variables is symplectic, the latter is

$$M_{gas} = 16\pi^3 L_0 \int_0^{J_{max}} \mathcal{F}_{0,eq}\, dJ. \tag{57}$$

The steady state is computed by an iteration which starts with $\Phi_{gas} = 0$. Given Φ_{gas} on a radial grid with spacing 0.01, we tabulate the function $J = A(E, L_0)/2\pi$ belonging to the potential $\Phi_{ext} + \Phi_{gas}$ by numerical quadrature (see Appendix A), which yields $E_{min}(L_0)$, the cutoff energy E_0 and, by inversion, the function $E(J, L_0)$. Next, we determine α from Eq. (57), compute the density from the second equation in (12), and obtain a new potential from Eq. (7). These steps are repeated until the maximum change of Φ_{gas} between two consecutive iterations is smaller than 10^{-13}. The iteration converges geometrically, more slowly for larger masses; for the polytropes in the Kepler potential with $0.01 \leq M_{gas}/M \leq 1$, for instance, between 6 and 42 iterations are required.

C. Initial data

The initial data considered in these notes are of two kinds: a gas cloud which is concentrated about one orbital phase and is far from any steady state, and a self-consistent steady state with a small perturbation. In both cases the initial DF is given as a function of the action-angle variables (Q, J) of a time-independent potential Φ_0, which is the external potential in the first case and the total potential Φ_{eq} of the steady state in the second one.

If the radii and momenta of the computational particles were drawn at random from the initial DF, the observables would be affected by fluctuations of the order of $N^{-1/2}$. For the numbers of particles used in this work, these fluctuations would exceed the values of $|h_n|/h_0$ which are reached during the mixing of the cloud, as well as the signal which is produced by the perturbation of the steady state. For this reason we place the computational particles on a regular lattice in the variables (Q, J),

$$J_a := \left(a - \frac{1}{2}\right) \frac{J_{max}}{N_J}, \quad a = 1, \ldots, N_J, \qquad Q_b := \left(b - \frac{1}{2}\right) \frac{2\pi}{N_Q}, \quad b = 1, \ldots, N_Q, \tag{58}$$

where $[0, J_{max}]$ is the range of the action variable in the support of the DF, such that $N = N_J N_Q$. The radii and momenta of the particles are obtained by inverting the map $(r, p_r) \mapsto (Q, J)$ which belongs to Φ_0, explicitly for the external potential (see below) and numerically for Φ_{eq} (see Appendix A), and the particle at the lattice point (Q_b, J_a) is given a mass m_{ab} which is proportional to the value of the initial DF at this point, with a constant of proportionality which is determined by the total mass M_{gas}. Since the map $(r, p_r) \mapsto (Q, J)$ is symplectic, all the cells of the lattice have the same volume in phase space. Hence, the sums over the computational particles in Eqs. (52,53) are midpoint rules for the corresponding integrals over phase space, and they are free of the fluctuations of a random sample. In plasma physics, a loading of the particles which avoids these fluctuations is known as a quiet start [45]. The method is only reliable over a finite time span: in the absence of self-gravity the rows of the lattice produce a spurious recurrence of the functions h_n, and in its presence the regularity of the lattice is eventually lost due to the discreteness of the particles. Both effects are quantified in section V.

As an initial datum far from equilibrium we consider the gas cloud

$$\mathcal{F}_0(0, Q, J) = \alpha \exp\left[-\frac{\sin^2(Q/2)}{\sigma_Q^2}\right] J^2 \exp\left(-\frac{J^2}{\sigma_J^2}\right), \tag{59}$$

where (Q, J) are the action-angle variables of the external potential and the constant $\alpha > 0$ is determined by the mass of the gas. For $\sigma_Q \ll 1$ the gas particles are concentrated near the pericenters $Q = 0$ of their orbits, and their action variables are distributed about $J = \sigma_J$, see Fig. 1. The factor J^2 makes the DF smooth at the circular orbit $J = 0$, where the angle variable is not defined. In the absence of self-gravity, the mixing of this cloud is the test problem of section V A, and in section VII we analyze how its evolution changes when the mass of the gas is increased. We truncate the DF at $J_{max} = 6\sigma_J$, where it has decayed to about 2×10^{-14} of its maximum. Unless otherwise stated,

$\sigma_Q = \sigma_J = 0.1$ and $N_J \times N_Q = 400 \times 25$. For the isochrone and the Kepler potentials the inversion of the map $(r, p_r) \mapsto (Q, J)$ only requires the solution of an equation of Kepler's type (see section II D), which we solve with Newton's method. In the Kepler case, for instance,

$$r = a(1 - e\cos\eta), \qquad p_r = \frac{\sqrt{a}\, e \sin\eta}{r}, \qquad Q = \eta - e\sin\eta, \tag{60}$$

with $a := (J + L_0)^2$ and $e := (1 - L_0^2/a)^{1/2}$. Since the cloud is not close to any steady state, its evolution is not compared with a reference simulation, and the observables are computed with the action-angle variables of the instantaneous potential, see section II E.

Initial data close to a self-consistent steady state are of the form

$$\mathcal{F}_0(0, Q, J) = \mathcal{F}_{0,eq}(J)\,[1 + \varepsilon\, s(J)\cos Q]\,, \qquad s(J) := \left(\frac{J}{J_{max}}\right)^{3/2}, \tag{61}$$

where (Q, J) are the action-angle variables which belong to Φ_{eq}, J_{max} is the action variable of the outermost orbit of the steady state, and $0 \le \varepsilon \le 1$ is the amplitude of the perturbation. The perturbation increases the density of the gas near the pericenters of the orbits and decreases it near their apocenters. It changes neither the mass of the gas nor the angle-average of its DF, and the factor $s(J)$ makes it regular at the circular orbit. On the lattice (58) the particle at the point (Q_b, J_a) has the mass

$$m_{ab} = c\,\mathcal{F}_{0,eq}(J_a)\,[1 + \varepsilon\, s(J_a)\cos Q_b]\,, \tag{62}$$

with a constant c which is determined by M_{gas}, such that the perturbation is encoded in the masses of the computational particles instead of their positions. Since $\sum_b \cos Q_b = 0$, the perturbation does not alter the mass which is assigned to each orbit. Particle methods in which the perturbation is carried by the weights of the particles have been used in Refs. [50, 51]. We stress that the lattice has to be constructed in the action-angle variables of the self-consistent potential Φ_{eq}: if those of the external potential are used instead, the resulting configuration is not stationary unless the mass of the gas is negligible.

Each simulation with $\varepsilon > 0$ is accompanied by a reference simulation with $\varepsilon = 0$ which uses the same lattice and the same numerical parameters, and the response of the gas to the perturbation is defined by

$$\chi_n(t) := \frac{h_n^{(\varepsilon)}(t) - h_n^{(0)}(t)}{\varepsilon\, h_0^{(0)}(0)}, \qquad \delta\Phi_{res}(t, r) := \frac{\delta\Phi^{(\varepsilon)}(t, r) - \delta\Phi^{(0)}(t, r)}{\varepsilon}, \tag{63}$$

where the superscripts refer to the amplitude of the perturbation. The subtraction of the reference simulation removes the errors which are common to both simulations, in particular the small departure of the discretized steady state from stationarity, which is quantified in section V. As long as the evolution is in the linear regime, the quantities (63) are independent of ε, and they can be compared directly with the solution of the linearized equations.

D. Numerical solution of the linearized equations

In order to compare the PIC simulations with the predictions of the linearized theory, we solve Eqs. (36,37) for the reduced model numerically as an initial-value problem. The function $\delta\mathcal{F}_0(t, Q, J)$ is represented by its values on a lattice of the form (58), whose points have the fixed radii $r_{ab} := r(Q_b, J_a)$, and it is advanced in time with the second-order splitting method of Strang [62]. Each time step consists of the update

$$\delta\mathcal{F}_0 \mapsto \delta\mathcal{F}_0 + \frac{\Delta t}{2}\frac{\partial\mathcal{F}_{0,eq}}{\partial J}\frac{\partial\delta\Phi}{\partial Q}, \tag{64}$$

followed by the exact solution of the transport equation $\partial_t\delta\mathcal{F}_0 + \Omega(J)\partial_Q\delta\mathcal{F}_0 = 0$ over the time Δt, which multiplies the n-th Fourier coefficient of $\delta\mathcal{F}_0$ with respect to Q by the factor $e^{-in\Omega(J)\Delta t}$, and by a second update of the form (64). The derivative with respect to Q is computed with the fast Fourier transform. The potential is obtained from the midpoint rule for Eq. (37),

$$\delta\Phi(t, r_{ab}) = -8\pi^2 G L_0\, \Delta Q\, \Delta J \sum_{c=1}^{N_J}\sum_{d=1}^{N_Q} \frac{\delta\mathcal{F}_0(t, Q_d, J_c)}{\max\{r_{ab}, r_{cd}\}}, \tag{65}$$

with $\Delta Q := 2\pi/N_Q$ and $\Delta J := J_{max}/N_J$, which is the exact potential of a set of concentric shells. Since the radii r_{ab} do not change in time, the lattice points need to be sorted only once, and the evaluation of Eq. (65) at all of them requires a number of operations proportional to $N_J N_Q$. Unless otherwise stated we use $N_J = 1600$, $N_Q = 32$ and $\Delta t = 0.5$.

The initial datum is $\delta\mathcal{F}_0 = \mathcal{F}_{0,eq}(J)s(J)\cos Q$, which corresponds to Eq. (61) with $\varepsilon = 1$, and the output consists of $\delta\Phi$ and of the functions

$$\chi_n(t) = \frac{\sum_{a,b} \delta\mathcal{F}_0(t, Q_b, J_a) B(J_a) e^{-inQ_b}}{\sum_{a,b} \mathcal{F}_{0,eq}(J_a) B(J_a)}, \tag{66}$$

which are the counterparts of the quantities defined in Eq. (63). This method is free of particle noise, but by construction it is restricted to the linear regime.

E. Computation of the gain of the self-consistent loop

For the reduced model, the operator $S(\omega)$ defined in Eq. (42) is discretized as follows. The harmonics are truncated at $n \le n_{max}$. The integral over J' in Eq. (40) is approximated by a Gauss-Legendre rule in the variable $\sigma \in (0,1)$ defined by $J = J_{max}(1-\sigma^p)$, with nodes J_a and weights w_a, $a = 1, \ldots, N_J$. This substitution concentrates the nodes at the edge $J = J_{max}$ of the support, where R_n varies rapidly when ω is close to Ω_{min}. For a DF which vanishes like $(E_0 - E)^k$ at the edge, Eq. (46) shows that at $\omega = \Omega_{min}$ the integrand is proportional to $\sigma^{p(k-1)-1} d\sigma$ for $\sigma \to 0$. We choose $p = 2$ for $k \ge 3/2$ and $p = 1/(k-1)$ for $1 < k < 3/2$, such that the integrand is bounded, and $p = 4$ for $k \le 1$. Furthermore, in order to avoid the loss of significant digits for the nodes which lie very close to the edge, the differences $E_0 - E(J)$ and $\Omega(J) - \Omega_{min}$ are evaluated from the expansion of the function $E(J)$ about J_{max} in powers of $J_{max} - J = J_{max}\sigma^p$. The integrals over the angles in Eq. (41) are approximated by the midpoint rule with N_Q points. In order to avoid the evaluation of the kernel for all pairs of points, we assign the points of each orbit to an auxiliary radial grid $\varrho_1 < \varrho_2 < \ldots < \varrho_{N_\varrho}$ with the linear weights W_i associated with its points, which gives

$$M_{nn'}(J_a, J_{a'}) \approx -4\pi G L_0 \sum_{i,i'=1}^{N_\varrho} \frac{P_{(n,a)i}\, P_{(n',a')i'}}{\max\{\varrho_i, \varrho_{i'}\}}, \qquad P_{(n,a)i} := \frac{2\pi}{N_Q} \sum_{b=1}^{N_Q} \cos(nQ_b)\, W_i(r(Q_b, J_a)). \tag{67}$$

The gain $\lambda(\omega)$ is then computed as the largest eigenvalue of the symmetric matrix with the entries

$$S_{(n,a)(n',a')}(\omega) = -\sqrt{w_a |R_n(J_a;\omega)|}\; M_{nn'}(J_a, J_{a'})\; \sqrt{w_{a'} |R_{n'}(J_{a'};\omega)|}, \tag{68}$$

which has the dimension $n_{max} N_J$, and the frequency ω_d of an oscillating mode is obtained by solving Eq. (44) with a bracketing root finder. Unless otherwise stated we use $n_{max} = 6$, $N_J = 200$, $N_Q = 128$ and $N_\varrho = 2000$.

For steady states with $k > 1$ the result converges rapidly. Doubling each of these resolutions changes λ_{edge} by at most 3×10^{-6} for the Wilson model with $M_{gas}/M = 0.097$ in the isochrone potential, and by at most 2×10^{-4}, 8×10^{-5} and 8×10^{-5} for the polytropes with $k = 2$, 1.5 and 1.25 and $M_{gas}/M = 1$ in the Kepler potential, respectively; the largest changes are due to the truncation of the harmonics and to the number of points in Q, whereas doubling N_J changes λ_{edge} by less than 10^{-6}. For $k \le 1$, where λ_{edge} is infinite, $\lambda(\omega)$ can be computed up to a distance of $10^{-12}(\Omega_{max} - \Omega_{min})$ from the edge: for the polytropes with $k = 0.75$ and $k = 1$ its value at this distance changes by less than 2×10^{-5} in relative terms when N_J is doubled. In all these cases the frequency ω_d of the mode changes by less than 2×10^{-6} when any of the resolutions is doubled.

F. Frequencies and damping rates

The frequency ω and the damping rate γ of the response are determined from the time series of $\chi_1(t)$, sampled at equidistant times, in the same way for the PIC simulations and for the solutions of the linearized equations. Within a given time window we fit the samples by a sum of K complex exponentials,

$$\chi_1(t) \approx \sum_{l=1}^{K} c_l\, e^{-i\omega_l t - \gamma_l t}, \tag{69}$$

using the matrix pencil method [63], and we identify (ω, γ) with the pair (ω_l, γ_l) of the term with the largest amplitude $|c_l|$. We use $K = 3$, and we estimate the uncertainty of the fit from the largest deviation among the results which

are obtained with $K = 2$ and $K = 3$ for the given window and for the windows which are shortened by ten percent at either end. This quantity measures the uncertainty of the fit and not the numerical error of the underlying time series, which is assessed through the convergence tests of section V. For oscillating modes, we alternatively determine the frequency from the maximum of the periodogram of $\chi_1(t)$.

V. CODE VALIDATION

In this section we validate the numerical methods described in the previous section. We start with the mixing of the gas cloud (59) in the absence of self-gravity, for which the exact solution is known. Next, we show that the functions $h_n(t)$ need to be computed with the action-angle variables which belong to the total potential, and we determine the time span over which the quiet start of section IV C is reliable. Then, we analyze the dependency of the response of a steady state to a perturbation on the time step and on the amplitude of the perturbation, and we compare it with the solution of the linearized equations. Finally, we test the linearized solver and the computation of the gain $\lambda(\omega)$ against each other.

A. Mixing in the absence of self-gravity

When the last term in the second equation in (49) is omitted, the computational particles move independently from each other in the external potential, and the functions $h_n(t)$ are given by Eq. (31). This provides a test for the time integration, for the representation of the DF by the lattice (58) and for the computation of the observables. We consider the reduced model with $L_0 = 2$ in the isochrone potential and the gas cloud (59) with $\sigma_Q = \sigma_J = 0.1$ (see Fig. 1), and we choose the weight function $B(J) = J^2 \exp(-J^2/\sigma_J^2)$, that is, Eq. (54) with $J_1 = 0$ and $\sigma_1 = \sigma_J$.

The dependency of the DF (59) on the angle variable has been chosen such that it is smooth and 2π-periodic and such that its Fourier coefficients (30) are known explicitly. In fact, the identity $\sin^2(Q/2) = (1 - \cos Q)/2$ yields

$$\exp\left[-\frac{\sin^2(Q/2)}{\sigma_Q^2}\right] = e^{-z} e^{z\cos Q} = e^{-z} \sum_{n\in\mathbb{Z}} I_n(z) e^{inQ}, \qquad z := \frac{1}{2\sigma_Q^2}, \tag{70}$$

where I_n denotes the modified Bessel function of the first kind of order n, which satisfies $I_{-n} = I_n$. The second equality in Eq. (70) is the generating function of these functions, see Eq. (10.35.2) in Ref. [64]; equivalently, $I_n(z)$ is the n-th Fourier coefficient of the function $e^{z\cos Q}$ (Eq. (10.32.3) in Ref. [64]). Consequently, the Fourier coefficients of the DF (59) with respect to the angle variable are

$$\hat{F}_n(J) = \alpha\, e^{-z} I_n(z)\, J^2 \exp\left(-\frac{J^2}{\sigma_J^2}\right). \tag{71}$$

For $\sigma_Q = 0.1$ one has $z = 50$, and in this case $I_n(z)/I_0(z)$ differs from $e^{-n^2/(2z)}$ by less than one percent for $|n| \le 16$. Hence, the narrower the pulse is in the angle variable, the more harmonics it contains: here the amplitudes of those with $|n| \le 2\sqrt{z} \approx 14$ are larger than $e^{-2} \approx 0.14$ times the one of the harmonic $n = 0$. It follows from Eqs. (31,71) that $|h_n(0)|/h_0 = I_n(z)/I_0(z)$, which is equal to 0.990, 0.960, 0.913 and 0.851 for $n = 1, 2, 3, 4$, and that $h_n(t)$ is given by an integral over J, which we compute with the trapezoidal rule on 2×10^5 subintervals of $0 \le J \le 6\sigma_J$.

Table I shows the error

$$e_n := \frac{1}{h_0} \max_{0\le t\le 2000} \left|h_n(t) - h_n^{exact}(t)\right| \tag{72}$$

for different lattices and time steps. For $N_J \times N_Q = 400 \times 64$ and $\Delta t = 0.1$ it does not exceed 6×10^{-9} for $n \le 4$. In this case the error is dominated by the time integration. In fact, when the Yoshida integrator is replaced with the exact advance $Q_j \mapsto Q_j + \Omega(J_j)\Delta t$ of the angle variables of the particles, the error drops to less than 3×10^{-11}, and the difference with respect to this solution decreases by the factors 16.0, 16.0 and 15.4 each time Δt is halved between 0.2 and 0.025 (for $n = 1$), as expected for a fourth-order method.

The remaining rows of Table I show the effect of the lattice. Since the sum (53) is a midpoint rule for an integral whose integrand is smooth and, with respect to Q, periodic, the error decreases rapidly when N_J or N_Q increase, until it reaches the one of the time integration.

With respect to the angle variable the error of the lattice can be computed explicitly. On a lattice with N_Q points per row the harmonics $n + mN_Q$, $m \in \mathbb{Z}$, cannot be distinguished from the harmonic n, such that the sum over the

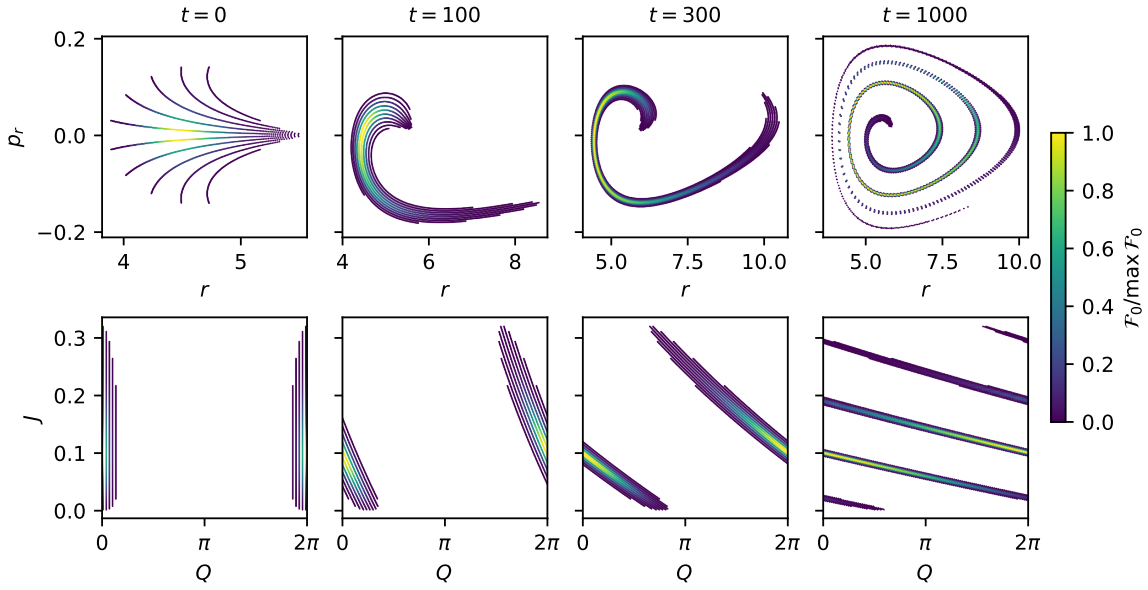


FIG. 1: Mixing of the initial datum (59) in the isochrone potential when self-gravity is neglected. Shown are the computational particles in the (r, p_r)-plane (upper panels) and in the plane of the action-angle variables (lower panels) at four different times, for a lattice with $N_J \times N_Q = 800 \times 64$ points. The color indicates the mass of each particle relative to the largest one, and only particles whose mass exceeds 10^{-3} of the latter are shown.

N_J	N_Q	Δt	e_1	e_2	e_3	e_4
50	64	0.1	4.1×10^{-6}	1.6×10^{-4}	0.15	0.72
100	64	0.1	3.3×10^{-8}	1.2×10^{-7}	5.1×10^{-7}	3.9×10^{-6}
200	64	0.1	5.3×10^{-9}	5.4×10^{-9}	5.9×10^{-9}	5.5×10^{-9}
400	64	0.1	5.3×10^{-9}	5.5×10^{-9}	6.0×10^{-9}	5.5×10^{-9}
400	16	0.1	9.2×10^{-3}	3.7×10^{-2}	8.2×10^{-2}	0.14
400	32	0.1	1.7×10^{-5}	7.3×10^{-5}	1.8×10^{-4}	3.7×10^{-4}
400	64	0.2	8.6×10^{-8}	8.8×10^{-8}	9.6×10^{-8}	8.8×10^{-8}
400	64	0.05	3.3×10^{-10}	3.4×10^{-10}	3.7×10^{-10}	3.4×10^{-10}
400	64	0.025	2.1×10^{-11}	2.0×10^{-11}	2.2×10^{-11}	2.7×10^{-11}
400	64	exact	6.6×10^{-12}	1.3×10^{-11}	2.0×10^{-11}	2.7×10^{-11}

TABLE I: Error e_n defined in Eq. (72) for the mixing of the initial datum (59) in the absence of self-gravity, for different lattices and time steps. In the last row the Yoshida integrator is replaced with the exact advance of the angle variables.

points Q_b yields $\sum_m (-1)^m \hat{F}_{n+mN_Q}$ instead of $\hat{F}_n$, where the signs are due to the fact that the Q_b are the midpoints of the subintervals. Since the transport does not mix different harmonics, this factor is independent of time. Taking into account that the masses of the computational particles are normalized such that their sum is equal to M_{gas}, Eq. (71) gives

$$e_n = \left| \frac{\sum_m (-1)^m I_{n+mN_Q}(z)}{\sum_m (-1)^m I_{mN_Q}(z)} - \frac{I_n(z)}{I_0(z)} \right|, \tag{73}$$

where the sums extend over all $m \in \mathbb{Z}$. This expression reproduces the values in the rows with $N_Q = 16$ and $N_Q = 32$ of Table I to nine significant digits, and for $N_Q = 64$ it is smaller than 5×10^{-15}. Hence, the number of points per row has to be a few times larger than the number $2\sqrt{z}$ of harmonics contained in the DF.

With respect to the action variable the situation is different, since for a given lattice the sum is only reliable over a finite time span. The reason is that the phases $n\Omega(J_a)t$ of two neighboring rows differ by $n|\Omega'(J)|\Delta J\, t$, with ΔJ the spacing between the rows. When this difference reaches 2π the rows are in phase again, such that h_n exhibits a spurious recurrence [45] at the time

$$t_{rec} = \frac{2\pi}{n|\Omega'(J)|\Delta J}. \tag{74}$$

In the present example, $\Delta J = 0.6/N_J$ and $|\Omega'(J)| \leq 3/c(L_0)^4 \approx 0.088$, which gives $t_{rec} \approx 120\, N_J/n$. For $N_J = 50$ this is 1980 and 1480 for $n = 3$ and $n = 4$, respectively, which explains the large values of e_3 and e_4 in the first row of the table. For the lattices which are used in the following subsections and in section VII the recurrence time for $n = 1$ exceeds the duration of the simulations: it is 4.7×10^4 for the one of section V B, at least 1.3×10^4 for those of sections V C and V D, and, with the frequencies of the external potential, at least 2.2×10^4 for those of section VII.

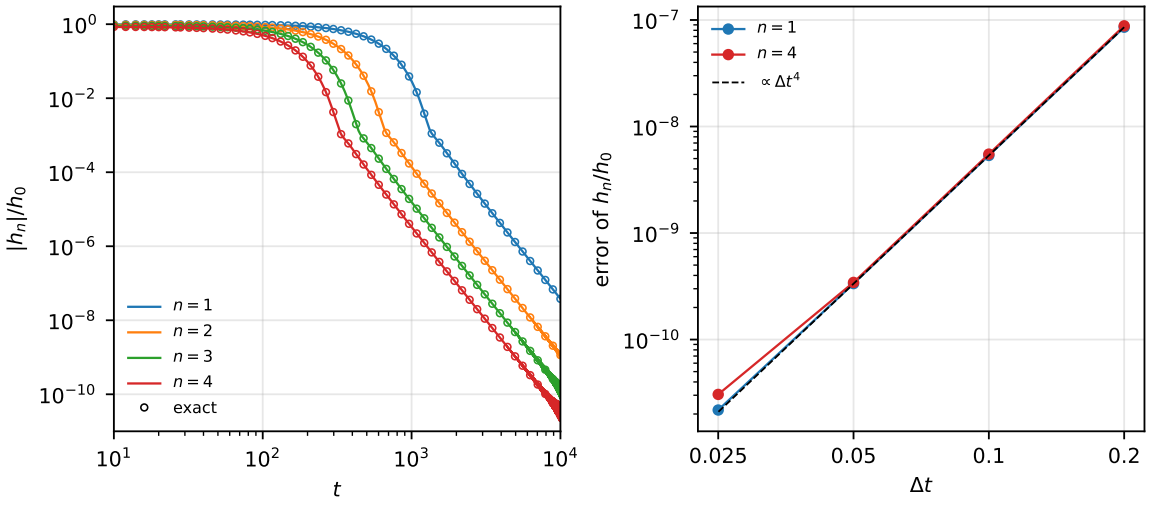


FIG. 2: Left panel: the functions $|h_n(t)|/h_0$, $n = 1, 2, 3, 4$, for the mixing of the initial datum (59) in the absence of self-gravity, obtained from the PIC code with $N_J \times N_Q = 800 \times 64$ and $\Delta t = 0.1$ (solid lines) and from the exact solution (31) (circles). Right panel: error of the time integration, measured through the difference with the solution in which the angle variables are advanced exactly, as a function of the time step. The dashed line is proportional to Δt^4.

Figure 2 shows the result of a simulation with $N_J \times N_Q = 800 \times 64$ and $\Delta t = 0.1$ which extends to $t = 10^4$. During this time $|h_1|$ and $|h_4|$ decay by seven and ten orders of magnitude, respectively, and the numerical solution reproduces the exact one over the whole range: for $n \leq 4$ the difference does not exceed $6 \times 10^{-9} h_0$ at any time, and it is smaller than $2 \times 10^{-10} h_0$ for $t \geq 5000$. At late times $|h_n(t)|$ is proportional to t^{-5} (a fit over $5000 \leq t \leq 10^4$ gives the exponents -5.03 and -5.01 for $n = 1$ and $n = 2$). This is the analogue of the algebraic tail (33) for the endpoint $J = 0$ of the support, at which the function $\hat{F}_n B$ vanishes like J^4. In the same simulation h_0, which is constant for the exact solution, varies by 1.1×10^{-9}, and the total energy by 7×10^{-11}, in relative terms.

B. Action-angle variables of the total potential

As emphasized in section II E, the functions $h_n(t)$ with $n \neq 0$ measure the departure of the gas configuration from a steady state only if they are computed with the action-angle variables which belong to the total potential. To illustrate this point we have repeated the simulation of the previous subsection including the self-gravity of the gas, for $M_{gas}/M = 10^{-4}$, 10^{-3} and 10^{-2}, with $N_J \times N_Q = 400 \times 25$, $\Delta r = 0.1$, $\Delta t = 0.1$ and up to $t = 2 \times 10^4$, using the weight function (54) with $J_1 = \sigma_1 = 0.1$. The cloud is far from a steady state, and the gas undergoes mixing while its potential Φ_{gas} readjusts: for $M_{gas}/M = 10^{-3}$ it changes by up to 40 percent of its largest value between $t = 0$ and $t = 2000$, and afterwards it is stationary up to fluctuations whose root mean square is smaller than 10^{-3} of this value.

Figure 3(a) shows the function $|h_1(t)|/h_0$ which is obtained with the action-angle variables of the external potential. When self-gravity is neglected it decays according to the law t^{-5} found in the previous subsection, until it reaches the level of the numerical errors, which is of the order of 10^{-9}, at $t \approx 10^4$. In contrast to this, when self-gravity is taken into account it settles down to a plateau: the magnitude of the time average of h_1/h_0 over the interval $2000 \leq t \leq 2 \times 10^4$ is 2.2×10^{-4}, 2.2×10^{-3} and 1.5×10^{-2} for the three values of M_{gas}/M, whereas it is 9×10^{-9} in the absence of self-gravity. For $M_{gas}/M = 10^{-3}$ the same value, 2.16×10^{-3}, is obtained with $N_J \times N_Q = 800 \times 25$ and 400×50. Such a plateau could be mistaken for a collective mode which is not damped.

However, the plateau disappears when the same particle data are analyzed with the action-angle variables of the potential $\Phi_{ext} + \langle \Phi_{gas} \rangle$, where $\langle \Phi_{gas} \rangle$ denotes the time average of the potential of the gas after it has become stationary. This is shown in Fig. 3(b) for $M_{gas}/M = 10^{-3}$: with these variables h_1/h_0 fluctuates about zero instead of about the

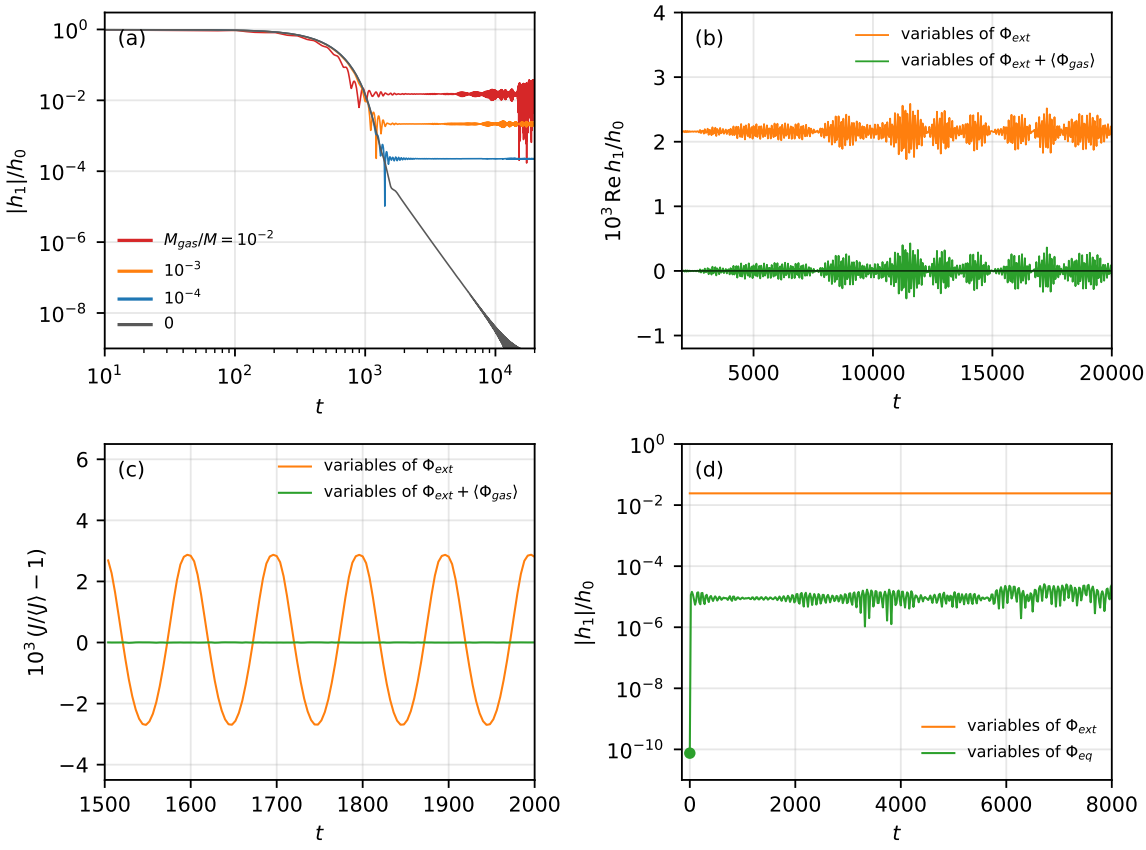


FIG. 3: Dependency of the function h_1 on the action-angle variables which are used to compute it. (a) $|h_1|/h_0$ for the mixing of the initial datum (59) including self-gravity, computed with the action-angle variables of the external potential, for different masses of the gas. (b) Real part of h_1/h_0 for $M_{gas}/M = 10^{-3}$, computed with the action-angle variables of the external potential and with those of the potential $\Phi_{ext} + \langle\Phi_{gas}\rangle$, where $\langle\Phi_{gas}\rangle$ is the average of Φ_{gas} over $t \geq 2000$. (c) Relative variation of the action variable of a single computational particle for the same mass, for both sets of variables; here $\langle\Phi_{gas}\rangle$ is the average over $1500 \leq t \leq 2000$, and $\langle J\rangle$ denotes the mean value over this interval. (d) $|h_1|/h_0$ for the reference simulation of the polytrope with $k = 2$ and $M_{gas}/M = 0.01$ in the Kepler potential, computed with the action-angle variables of the external potential and with those of Φ_{eq}.

value 2.16×10^{-3}, and the magnitude of its time average over $2000 \leq t \leq 2 \times 10^4$ is reduced to 5.4×10^{-7}, which is 4000 times smaller. The action-angle variables of the instantaneous potential, which we use for the initial data of section VII, give the same result: with them the magnitude of the time average is 1.5×10^{-6}. Therefore, the plateau is due to the choice of the coordinates and not to the dynamics. The reason is that the action variable of the external potential is not conserved along the trajectories of the gas particles in the total potential. As shown in Fig. 3(c), for a given particle it oscillates with the period of the radial motion; the relative difference between its largest and its smallest value is 5.6×10^{-3} for the particle shown, which has $J = 0.10$, and 4.4×10^{-3} for a particle with $J = 0.30$. In contrast to this, the action variable which belongs to the total potential is constant to within a relative variation of 2×10^{-5}. Consequently, a DF which depends only on the action variable of the total potential is not independent of the angle variable of the external one, the difference between both sets of variables being of the first order in M_{gas}/M.

What remains after subtracting the time average is a fluctuation which is the same for both sets of variables, see Fig. 3(b). Its root mean square is $2.0 \times 10^{-4} h_0$ in the simulation with 400×25 particles, and $1.1 \times 10^{-4} h_0$ and $9.6 \times 10^{-5} h_0$ in those with 800×25 and 400×50 particles, respectively. For $M_{gas}/M = 10^{-2}$ it grows in time and is visible in Fig. 3(a) for $t > 10^4$; in this case its root mean square is $9.1 \times 10^{-3} h_0$ with 400×25 particles and $2.2 \times 10^{-3} h_0$ with 800×25 particles. Hence, these fluctuations are caused by the finite number of particles, which is the subject of the next subsection.

The same occurs for a self-consistent steady state. Figure 3(d) shows the result for the reference simulation of the polytrope (55) with $k = 2$ and $M_{gas}/M = 0.01$ in the Kepler potential, which is analyzed in section VI, with the weight function (54) with $J_1 = 0.35$ and $\sigma_1 = 0.2$. The value of $|h_1|/h_0$ which is obtained with the action-angle variables of the external potential is 2.4×10^{-2} at all times, whereas with those of Φ_{eq} it is 7.5×10^{-11} at $t = 0$

and remains below 3×10^{-5} up to $t = 8000$. The small value which develops for $t > 0$ reflects the departure of the discretized steady state from stationarity, which is analyzed next.

C. Stationarity of the discretized steady states

Next, we analyze the reference simulations with $\varepsilon = 0$, in which the gas should remain in the self-consistent steady state. We consider the polytrope (55) with $k = 1.25$, $L_0 = 2$ and $J_{max} = 0.7$ in the Kepler potential with $M_{gas}/M = 1$, which is the largest mass ratio considered in these notes and, hence, the most demanding case, and the weight function (54) with $J_1 = 0.35$ and $\sigma_1 = 0.2$. We use two quantities to measure the departure from stationarity: $|h_1^{(0)}(t)|/h_0$, which vanishes for a steady state, and the root mean square displacement of the action variables of the computational particles,

$$\Delta J_{rms}(t) := \left[\frac{1}{M_{gas}} \sum_{j=1}^{N} m_j \left(J_j(t) - J_j(0) \right)^2 \right]^{1/2} , \tag{75}$$

both computed with the action-angle variables of Φ_{eq}. The results for three different numbers of particles and two values of Δr are shown in Fig. 4. They reveal two different effects.

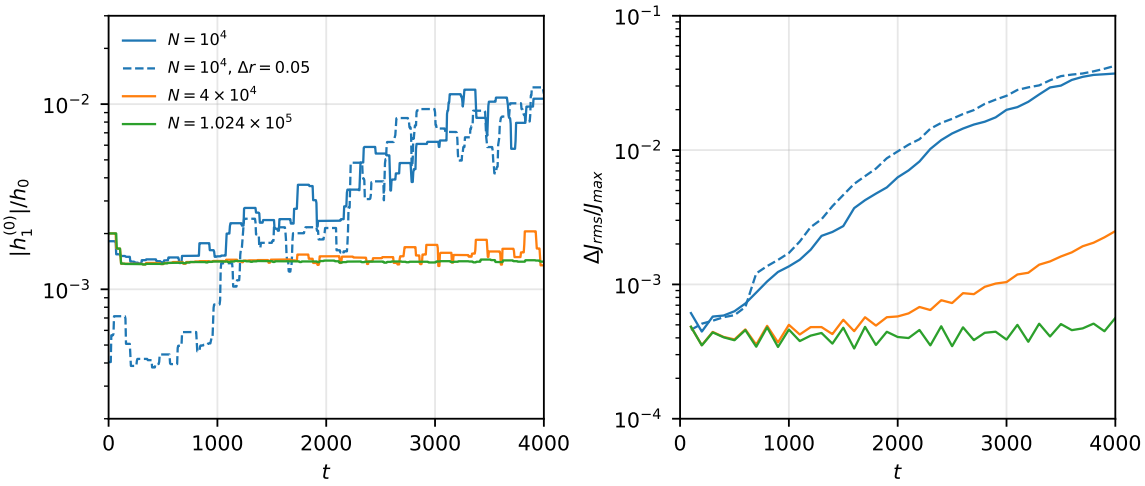


FIG. 4: Reference simulations ($\varepsilon = 0$) for the polytrope with $k = 1.25$ and $M_{gas}/M = 1$ in the Kepler potential, for different numbers $N = N_J N_Q$ of particles and grid spacings ($\Delta r = 0.1$ unless otherwise stated). Left panel: largest value of $|h_1^{(0)}|/h_0$ in a moving window of width 100. Right panel: root mean square displacement (75) of the action variables of the particles.

The first effect is present from the beginning. Within the first radial periods, $|h_1^{(0)}|/h_0$ rises to a level of about 1.4 to 2×10^{-3}, which does not depend on the number of particles and is reduced by a factor between 2.5 and 3.4 when Δr is halved. Its origin is the discretization error of the potential: the lattice represents a steady state of the exact potential Φ_{eq}, whereas the particles move in the potential computed on the grid, which differs from Φ_{eq} by an error of the order of Δr^2. Hence, the initial configuration is slightly out of equilibrium, and the gas reacts to this. For the Wilson model with $W_0 = 3$, $J_{max} = 0.138$ and $M_{gas}/M = 0.5$ in the isochrone potential we have verified that the resulting perturbation $\delta\Phi^{(0)}$ of the potential decreases by a factor of four when Δr is halved, whereas it changes by less than three percent when the number of particles is multiplied by four or the time step is halved. Since this error is the same in the simulation with the perturbation and in its reference simulation, to leading order it cancels in the response (63).

The second effect is a growth which sets in after a certain time. For $N = 10^4$ the quantity $|h_1^{(0)}|/h_0$ starts to grow at $t \approx 1000$ and reaches 10^{-2} at the end of the simulation, while the displacement of the action variables increases from $1.4 \times 10^{-3} J_{max}$ at $t = 1000$ to $3.7 \times 10^{-2} J_{max}$ at $t = 4000$. This growth is not reduced by halving Δr, but it is delayed when the number of particles is increased: for $N = 4 \times 10^4$ the displacement stays below $6 \times 10^{-4} J_{max}$ up to $t = 2000$ and reaches $2.5 \times 10^{-3} J_{max}$ at $t = 4000$, and for $N = 1.024 \times 10^5$ both quantities remain constant up to $t = 4000$. Therefore, it is due to the discreteness of the particles, which gradually destroys the regularity of the

lattice. Note that the dependency on N is much stronger than the one of the fluctuations of a random sample, which are proportional to $N^{-1/2}$. For a gas of small mass the lattice is preserved over much longer times: for the polytrope with $k = 2$ and $M_{gas}/M = 0.01$ and $N = 10^4$ we find $\Delta J_{rms} = 3.6 \times 10^{-5} J_{max}$ at $t = 8000$. We conclude that for $M_{gas}/M = 1$ the quiet start with $N = 10^4$ particles is reliable up to $t \approx 1000$, whereas $N \geq 4 \times 10^4$ particles are required for the time span $t \leq 4000$ of our simulations. In all these simulations the total energy is conserved to better than 10^{-5}.

Finally, we mention a test of the computation of the gravitational field in a situation which is dominated by self-gravity, namely the collapse of a cold homogeneous sphere of mass M_{gas} and radius R in the absence of an external potential. Since in this case all the particles have $L = 0$, this test has been performed with an implementation of the same algorithm for a general distribution of the angular momenta. Before the shells cross each other, the shell with initial radius r_0 follows the cycloid $r = r_0 \cos^2\theta$, $t = (\theta + \sin\theta\cos\theta)/\sqrt{2GM_{gas}/R^3}$. For $\Delta r = 0.01R$, the median of the relative error of the radii of the shells with $0.1 < r_0/R < 0.9$ at the time $t = 0.8\sqrt{R^3/(GM_{gas})}$ is 1.8×10^{-5} for $N = 200$ and 4.3×10^{-6} for $N = 400$ shells, which corresponds to second-order convergence in $1/N$.

D. Time step, amplitude of the perturbation and comparison with the linearized equations

We now turn to the response (63) of the gas to a perturbation of the form (61). We first analyze its dependency on the time step, for the polytrope with $k = 1$ and $M_{gas}/M = 1$ in the Kepler potential (with the remaining parameters as in the previous subsection), $\varepsilon = 0.1$, $N_J \times N_Q = 400 \times 25$ and $\Delta r = 0.1$, by repeating the simulation and its reference simulation with $\Delta t = 0.2$, 0.1 and 0.05. We measure the difference between the responses $\delta\Phi_{res}$ obtained with different time steps with the norm (28) for $r_a = 3$ and $r_b = 7.5$, relative to the largest value of $\|\delta\Phi_{res}\|$ in the intervals $0 \leq t \leq 250$, $250 \leq t \leq 500$, $500 \leq t \leq 750$ and $750 \leq t \leq 1000$. The difference between the results for $\Delta t = 0.1$ and $\Delta t = 0.05$ grows from 6.6×10^{-4} in the first interval to 1.9×10^{-2} in the last one, and the one between $\Delta t = 0.2$ and $\Delta t = 0.05$ is larger by factors between 4.4 and 6.4. For a second-order method one expects the factor five, and for a fourth-order one the factor 17. Hence, in the presence of self-gravity the time integration is only second-order accurate, in contrast to the fourth-order convergence found in section V A. This reduction of the order is to be expected, since the field which results from the linear weights (50) is continuous but not differentiable with respect to the positions of the particles, whereas the fourth-order accuracy of the composition method requires a smooth force. At later times the differences between the simulations grow to several ten percent, as a consequence of the loss of the quiet start at $t \approx 1000$ found in the previous subsection: at $t = 2000$ the radii of corresponding particles in the simulation with $\Delta t = 0.1$ and in a further one with $\Delta t = 0.025$ differ by up to 3.3. In contrast to this, the frequency of the oscillation of $\delta\Phi_{res}$, determined from its periodogram over $500 \leq t \leq 2000$, is 0.14770 for the three time steps, and the relative error of the total energy, 1.8×10^{-5}, does not depend on Δt.

Next, we compare the response with the solution of the linearized equations obtained with the method of section IV D. For a gas of small mass we consider the polytropes with $k = 0.75$, 1, 1.5 and 2 and $M_{gas}/M = 0.01$ in the Kepler potential, with $N_J \times N_Q = 400 \times 25$, $\Delta r = 0.1$ and $\Delta t = 0.1$. In these cases the response decays algebraically (see section VI), and at $t = 1000$ the magnitude of χ_1 has decreased to between 2.6×10^{-3} (for $k = 0.75$) and 1.8×10^{-4} (for $k = 2$) of its initial value. With $\varepsilon = 0.1$, the difference between the functions $\chi_1(t)$ obtained from the PIC code and from the linearized equations is smaller than $1.2 \times 10^{-4}|\chi_1(0)|$ for $0 \leq t \leq 2000$ and the four values of k. Relative to the largest value of $|\chi_1|$ in the respective interval, it is at most 1.2×10^{-4} for $0 \leq t \leq 500$ and between 0.3 and 0.5 percent for $500 \leq t \leq 1000$. With $\varepsilon = 0.5$ the corresponding numbers are 5×10^{-4} and 1.4 percent. Therefore, for this mass ratio the two methods, which are based on entirely different discretizations, agree with each other to better than one percent while the response decays by more than two orders of magnitude, and the dependency on the amplitude of the perturbation is weak.

For $M_{gas}/M = 1$ the situation is different, since the response depends on ε even for rather small amplitudes. We consider again the polytrope with $k = 1.25$, for which the linearized equations predict an oscillating mode (see section VI), and compare three simulations with $\varepsilon = 0.1$, 0.03 and 0.01 which share the lattice with $N_J \times N_Q = 1280 \times 80$ points, the time step $\Delta t = 0.05$ and the reference simulation, such that they only differ from each other in the amplitude. In order to compare two oscillating signals we use the quantity

$$\kappa(t_0) := \frac{\sum_j w(t_j - t_0)\, \chi_1(t_j)\, \chi_1^{lin}(t_j)^*}{\sum_j w(t_j - t_0)\, |\chi_1^{lin}(t_j)|^2}, \qquad w(\tau) := \cos^2\left(\frac{\pi\tau}{T_w}\right) \quad \text{for } |\tau| \leq \tfrac{T_w}{2}, \tag{76}$$

and $w(\tau) := 0$ otherwise, where χ_1^{lin} refers to the solution of the linearized equations, the sums extend over the output times t_j which are common to both solutions, and $T_w = 200$, which corresponds to about five periods of the oscillation. The magnitude of $\kappa(t_0)$ is the ratio between the amplitudes of the two signals in the window centered at

t_0, and its phase is their phase difference, such that $-d\arg\kappa/dt_0$ is the difference between their frequencies. Since κ depends linearly on χ_1, it can be extrapolated to $\varepsilon = 0$.

The results are shown in Fig. 8(a) of section VI E, as functions of $\omega_b t_0$. For $\varepsilon = 0.1$ the amplitude of the response falls to 56 percent of the one predicted by the linearized equations at $t_0 = 1300$, and for $\varepsilon = 0.03$ to 91 percent at $t_0 = 2000$, whereas for $\varepsilon = 0.01$ it stays between 0.977 and 1.005 of the latter for $t_0 \leq 2000$ and between 0.95 and 1.06 up to $t_0 = 3900$. A simulation with $\varepsilon = 0.1$, $N = 10^4$ and $\Delta t = 0.1$ agrees with the one with $N = 1.024 \times 10^5$ to within two percent for as long as its quiet start lasts. Hence, the loss of amplitude is not a numerical artifact but a nonlinear effect, whose origin is discussed in section VI E. When the three results are extrapolated to $\varepsilon = 0$, by a straight line through the two smallest amplitudes or by a parabola, the amplitude lies between 0.99 and 1.06 times the linear one for $t_0 \leq 2000$. The phase of κ shows that the frequency of the oscillation decreases slightly with the amplitude, and that its extrapolation to $\varepsilon = 0$ differs from the frequency of the linearized solution by at most 2.5×10^{-5}, which is of the order of the accuracy with which the latter is known (see the next subsection). Note that for $\varepsilon = 0.01$ the function $h_1^{(0)}/\varepsilon$ of the reference simulation is 2.5 times larger than the response itself, such that the latter is obtained as the difference of two much larger quantities.

We conclude that for $M_{gas}/M = 1$ the PIC code reproduces the solution of the linearized equations within a few percent, provided that $N \approx 10^5$ particles and amplitudes $\varepsilon \leq 0.01$ are used. [(Erik) For $k = 0.75$, 1 and 2 the simulations of section VI with $M_{gas}/M = 1$ still have $N = 10^4$ particles and $\varepsilon = 0.1$. The ones with $N = 1.024 \times 10^5$, $\Delta t = 0.05$ and $\varepsilon = 0.03$ and 0.1 have been run on the cluster and their analysis is pending; the default parameters quoted at the end of section IV A have to be adapted accordingly.]

E. Linearized equations and gain of the self-consistent loop

In order to estimate the accuracy of the solutions of the linearized equations, we have repeated the computation for the polytrope with $k = 1.25$ and $M_{gas}/M = 1$ of the previous subsection with half the time step, with twice the number of points in Q and with twice the number of rows in J, and compared the results with the one for the default values $N_J \times N_Q = 1600 \times 32$ and $\Delta t = 0.5$ by means of the quantity (76) in the interval $300 \leq t_0 \leq 3900$. Halving Δt changes the amplitude of χ_1 by 0.2 percent and its frequency by -0.9×10^{-5}, doubling N_Q changes them by -0.3 percent and 1.4×10^{-5}, and doubling N_J by 0.01 percent and 0.7×10^{-6}, respectively. Since the differences in the frequency accumulate in the phase, the difference between the functions $\chi_1(t)$ themselves grows linearly in time; at $t = 4000$ it reaches 3.5, 5.9 and 0.3 percent of the amplitude in the three cases.

An independent test is provided by the gain $\lambda(\omega)$, which is computed by a completely different method (see section IV E). For the polytropes with $k = 0.75$, 1 and 1.25 and $M_{gas}/M = 1$ in the Kepler potential, the solution of the linearized equations settles down to an oscillation whose frequency, determined with the method of section IV F in the window $300 \leq t \leq 1500$, is 0.14169, 0.14783 and 0.15330, respectively, whereas the equation $\lambda(\omega_d) = 1$ yields $\omega_d = 0.14169$, 0.14785 and 0.15326. Hence, both methods agree to within 4×10^{-5}.

VI. INITIAL DATA CLOSE TO A STEADY STATE

In this section we consider initial data of the form (61), that is, small perturbations of a self-consistent steady state, whose evolution can be compared with the solutions of the linearized equations and with the predictions of the gain $\lambda(\omega)$. We start with the two families of steady states and with the threshold mass above which they possess an oscillating mode. Then, we analyze the response of the gas below the threshold, where it decays, and above it, where it settles down to an oscillation which is not damped. Finally, we show that at finite amplitude the gas particles near the edge of the steady state are trapped by the oscillating mode, and we estimate the amplitude above which this occurs.

A. Steady states

We consider the two families of self-consistent steady states of section IV B, in both cases with $L_0 = 2$. The first one consists of the lowered Maxwellians (56) with $W_0 = 3$ in the isochrone potential, with $g = 2$ and $J_{max} = 0.138$ unless otherwise stated and $0.003 \leq M_{gas}/M \leq 0.5$. The second one consists of the polytropes (55) in the Kepler potential with $k = 0.75$, 1, 1.25, 1.5, 2 and 3, $J_{max} = 0.7$ and $0.01 \leq M_{gas}/M \leq 1$, for which the dichotomy of Ref. [31] has been established. (In the Kepler case the value of L_0 can be fixed by means of the rescaling (14), such that this family is characterized by k, J_{max}/L_0 and M_{gas}/M.) The two families differ in the width of their frequency bands: in the limit $M_{gas}/M \to 0$ the band is $[0.0602, 0.0711]$ for the first one, whose width is only 17 percent of its

mean value, and $[0.0508, 0.125]$ for the second one. The properties of the steady states are summarized in Table II. As the mass of the gas increases, the gas contracts and the frequencies of all the orbits grow. In all the cases the DF is a decreasing function of the energy and $\lambda(0) < 1$, such that the steady states are linearly stable with respect to spherically symmetric perturbations (see section III C). Furthermore, the second largest eigenvalue of the operator $S(\omega)$ is smaller than 0.26 up to the edge of the band, such that there is at most one oscillating mode with frequency below the band.

	M_{gas}/M	r_{in}	r_{out}	Ω_{min}	Ω_{max}	$\lambda(0)$	λ_{edge}
lowered Maxwellian, $g = 2$	0.0065	4.31	8.50	0.06097	0.07230	0.018	0.100
	0.042	4.25	8.29	0.06548	0.07915	0.105	0.555
	0.058	4.23	8.20	0.06756	0.08230	0.140	0.720
	0.075	4.20	8.11	0.06979	0.08570	0.174	0.874
	0.097	4.17	7.99	0.07271	0.09017	0.215	1.047
	0.17	4.06	7.63	0.08275	0.10552	0.326	1.468
	0.5	3.60	6.40	0.13419	0.18434	0.586	2.210
polytrope, $k = 0.75$	0.01	2.39	12.10	0.05159	0.12618	0.008	∞
$k = 1$	0.01	2.39	12.10	0.05161	0.12626	0.008	∞
$k = 1.5$	0.01	2.39	12.10	0.05163	0.12640	0.009	0.018
$k = 2$	0.01	2.39	12.09	0.05165	0.12652	0.010	0.016
polytrope, $k = 0.75$	1	1.83	7.57	0.15159	0.28367	0.387	∞
$k = 1$	1	1.81	7.50	0.15399	0.29366	0.402	∞
$k = 1.25$	1	1.80	7.45	0.15601	0.30303	0.415	1.583
$k = 1.5$	1	1.79	7.40	0.15774	0.31186	0.428	1.051
$k = 2$	1	1.78	7.32	0.16053	0.32814	0.452	0.820
$k = 3$	1	1.75	7.21	0.16434	0.35648	0.492	0.738

TABLE II: Properties of the self-consistent steady states of the reduced model whose perturbations are analyzed in this section: the lowered Maxwellians (56) with $W_0 = 3$ and $J_{max} = 0.138$ in the isochrone potential, and the polytropes (55) with $J_{max} = 0.7$ in the Kepler potential, in both cases with $L_0 = 2$. Shown are the inner and outer radii of the shell which is occupied by the gas, the frequency band, and the gain of the self-consistent loop at zero frequency and at the edge of the band.

In the remainder of this section the response of the gas to the perturbation (61) is described by the function $\chi_1(t)$ defined in Eq. (63), which is computed with the weight function (54) with $J_1 = \sigma_1 = 0.05$ for the lowered Maxwellians and with $J_1 = 0.35$ and $\sigma_1 = 0.2$ for the polytropes. With these choices $|\chi_1(0)| = 0.138$ in the first case and $0.17 \leq |\chi_1(0)| \leq 0.22$ in the second one. The PIC simulations use $N_J \times N_Q = 400 \times 25$ and $\Delta r = 0.1$, with $\Delta t = 0.05$ for the lowered Maxwellians and $\Delta t = 0.1$ for the polytropes, except for the polytrope with $k = 1.25$ and $M_{gas}/M = 1$, for which we use the simulations with $N_J \times N_Q = 1280 \times 80$ and $\Delta t = 0.05$ of section V D.

B. Gain of the self-consistent loop and threshold mass

We have computed the gain $\lambda(\omega)$ with the method of section IV E for all the steady states of the previous subsection. We measure the distance of a frequency $\omega < \Omega_{min}$ from the edge of the band in units of the width of the band,

$$\delta := \frac{\Omega_{min} - \omega}{\Omega_{max} - \Omega_{min}}, \tag{77}$$

and we denote by δ_d the value of this quantity for the frequency ω_d of an oscillating mode.

Figure 5(c) confirms the behavior of the gain at the edge of the band which was derived in section III D: for $k > 1$ it converges to a finite value λ_{edge} when $\delta \to 0$, whereas for $k \leq 1$ it grows without bound. In the interval $10^{-8} \leq \delta \leq 10^{-5}$ the local exponent $d\log(\lambda_{edge} - \lambda)/d\log\delta$ differs from $\min\{k-1, 1\}$ by at most 0.01 for $k = 1.25$, 1.5 and 3, and for $k = 2$ it lies between 0.92 and 0.95, which reflects a logarithmic correction; for $k = 0.75$ and $k = 1$ the gain grows like δ^{k-1} and $|\log\delta|$, respectively. Figures 5(a) and (b) show that λ_{edge} increases monotonically with the mass of the gas, proportionally to M_{gas}/M for small masses, and reaches the value one at a threshold mass. We have determined this mass by solving the equation $\lambda_{edge} = 1$ with a bracketing root finder, computing a self-consistent steady state for each trial value of the mass. The results are given in Table III.

The threshold is much smaller for the lowered Maxwellians than for the polytropes, and within the first family it grows with J_{max}. Both observations can be attributed to the width of the frequency band: the narrower the band, the smaller the perturbation of the potential which is required in order to keep the orbits in phase. A threshold of

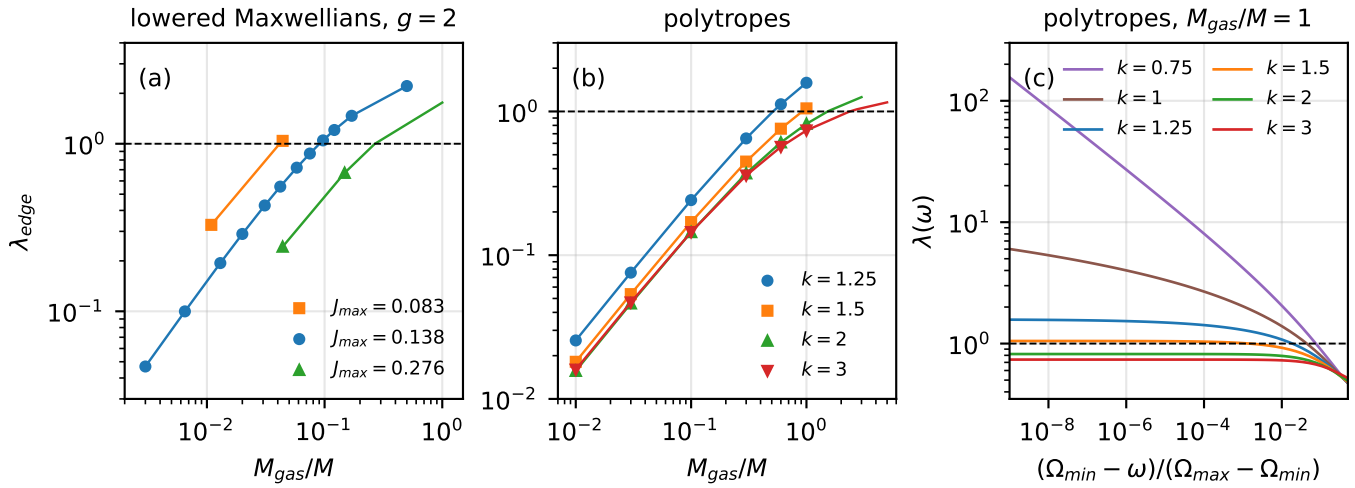


FIG. 5: Gain of the self-consistent loop. (a) The value λ_{edge} at the edge of the frequency band as a function of the mass of the gas, for the lowered Maxwellians with $g = 2$ in the isochrone potential and three values of J_{max}. (b) The same for the polytropes with $k > 1$ in the Kepler potential. The symbols refer to the steady states of section VI A, and the lines include those which were computed in order to determine the threshold mass. (c) The gain as a function of the distance (77) from the edge of the band, for the polytropes with $M_{gas}/M = 1$. An oscillating mode exists if and only if the curve crosses the dashed line $\lambda = 1$.

		threshold M_{gas}/M	Ω_{min}	Ω_{max}
lowered Maxwellian, $g = 2$	$J_{max} = 0.083$	0.0416	0.0705	0.0807
	$J_{max} = 0.138$	0.0906	0.0719	0.0889
	$J_{max} = 0.276$	0.268	0.0794	0.1150
polytrope	$k = 1.25$	0.514	0.0989	0.2057
	$k = 1.5$	0.920	0.1473	0.2935
	$k = 2$	1.53	0.2390	0.4737
	$k = 3$	2.38	0.4095	0.8526

TABLE III: Threshold mass, defined by the condition $\lambda_{edge} = 1$, above which an oscillating mode with frequency below the band exists, and frequency band of the steady state at the threshold. For the lowered Maxwellians with $g \leq 1$ and for the polytropes with $k \leq 1$, λ_{edge} is infinite and a mode exists for any mass.

this kind was found by Louis [65] for sequences of one-dimensional polytropes and King models. For the polytropes in the Kepler potential, the results of Ref. [31] hold when the mass of the gas is sufficiently small, and Table III shows how small it has to be: for $k = 1.25$, for instance, the perturbations can only be damped as long as $M_{gas}/M < 0.514$. At a fixed mass, the gain is larger when the DF is less regular at the edge of its support; for the polytropes with $M_{gas}/M = 1$, λ_{edge} increases from 0.738 for $k = 3$ to 1.583 for $k = 1.25$. This is the quantitative counterpart of the observation of Ref. [39] that a sharp edge favors the existence of an oscillating mode.

When $\lambda_{edge} > 1$, the frequency of the mode follows from Eq. (44). Slightly above the threshold the mode lies just below the band, as for the modes of the one-dimensional slab [39, 65], and as the mass increases it separates from the band (see Table IV below). For $k \leq 1$ an oscillating mode exists for any mass, but when the mass is small its frequency lies extremely close to Ω_{min}, as anticipated in Eq. (47): for the polytrope with $k = 0.75$ and $M_{gas}/M = 0.01$ we find $\delta_d = 2.0 \times 10^{-8}$, and for $k = 1$ and the same mass the extrapolation of the logarithmic law yields $\delta_d \approx 10^{-98}$. Hence, for the steady states with $k \leq 1$ and small mass, to which the existence result of Ref. [31] refers, the oscillating mode is present but cannot be resolved in a time evolution: for $k = 0.75$ and $M_{gas}/M = 0.01$ this would require the time $2\pi/(\Omega_{min} - \omega_d) \approx 4 \times 10^9$. A similar difficulty in classifying, by means of time evolutions, the steady states which lie close to the threshold is reported in Ref. [34] for isotropic polytropes.

C. Decay of the perturbation below the threshold

We now turn to the time evolution of the perturbation (61), starting with the steady states for which $\lambda_{edge} < 1$.

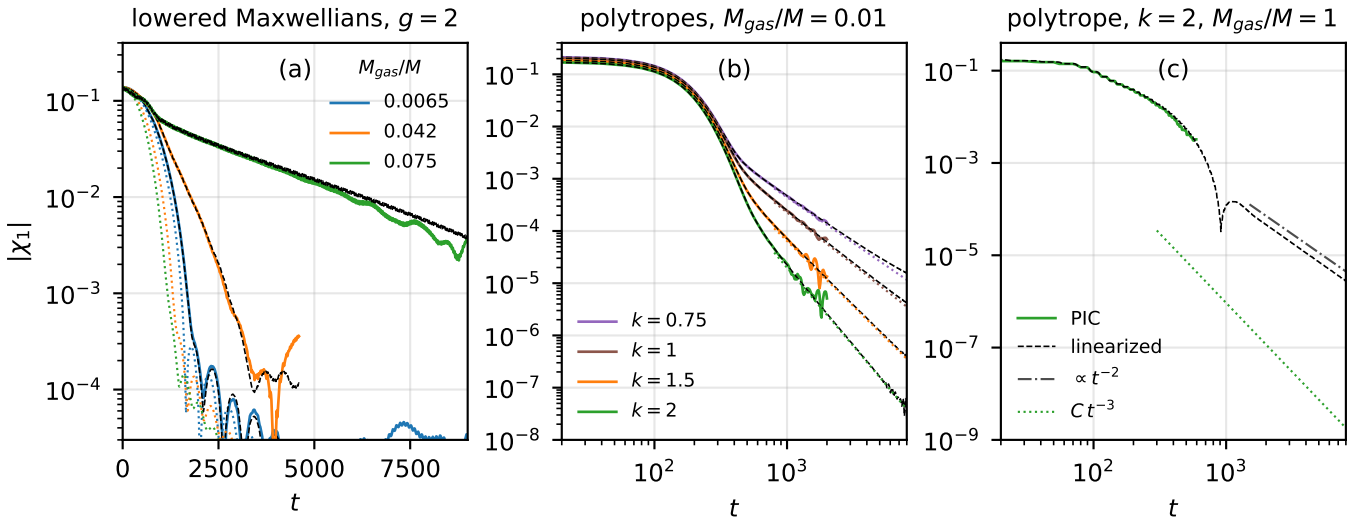


FIG. 6: Response of the gas to the perturbation (61) for steady states without oscillating mode. (a) Lowered Maxwellians with $g = 2$ and $J_{max} = 0.138$ below the threshold mass. The solid lines are the results of the PIC simulations (with $\varepsilon = 1$, 0.065 and 0.075 in the order of increasing mass), the dashed lines the solutions of the linearized equations, and the dotted lines the solutions which are obtained when the self-gravity of the perturbation is omitted. (b) Polytropes with $M_{gas}/M = 0.01$. The solid lines are the results of the PIC simulations with $\varepsilon = 0.1$, which are shown for $t \leq 2000$, the dashed lines the solutions of the linearized equations, and the dotted lines the asymptotic expression (78) for the mixing in a fixed potential. (c) Polytrope with $k = 2$ and $M_{gas}/M = 1$. Shown is the envelope of $|\chi_1|$ (largest value in a moving window of width 40) for the PIC simulation with $\varepsilon = 0.1$, up to $t = 600$, and for the solution of the linearized equations with $N_J = 3200$. The dash-dotted line is proportional to t^{-2}, and the dotted line is the expression (78).

Figure 6(a) shows the function $|\chi_1(t)|$ for the lowered Maxwellians with $M_{gas}/M = 0.0065$, 0.042 and 0.075, for which $\lambda_{edge} = 0.100$, 0.555 and 0.874, respectively. Besides the PIC simulations and the solutions of the linearized equations (36,37), it shows the solutions of Eq. (36) without its third term, which describe the mixing of the perturbation in the fixed potential Φ_{eq}. The self-gravity of the perturbation delays the mixing: the response falls below one percent of its initial value at $t = 1650$, 2650 and 11300 for the three masses, compared to $t = 1510$, 1280 and 1130 in the fixed potential. (A delay of the phase mixing by self-gravity has also been found in N-body simulations of the vertical motion in a galactic disk [66].) For $M_{gas}/M = 0.075$ the response is a damped oscillation over more than two orders of magnitude, which is the analogue of Landau damping discussed at the end of section III D. With the method of section IV F we obtain, in the interval $2000 \leq t \leq 4000$, $\omega = 0.07021 \pm 6 \times 10^{-5}$ and $\gamma = (3.4 \pm 0.2) \times 10^{-4}$ from the linearized equations, and $\omega = 0.07028 \pm 4 \times 10^{-6}$ and $\gamma = (3.48 \pm 0.06) \times 10^{-4}$ from the PIC simulation. The frequency lies inside the band, at 3 percent of its width above the lower edge, and for $t \gtrsim 1.4 \times 10^4$ the decay becomes slower, which marks the transition to an algebraic tail. For $M_{gas}/M = 0.058$ the frequency lies at 7 percent of the width of the band above the edge, and the damping rate is three times larger. Hence, as the threshold is approached from below, the frequency of the damped oscillation approaches the edge of the band and its damping rate tends to zero; above the threshold it becomes the oscillating mode of section VI D. This is the behavior described by Louis [65] for the one-dimensional slab, and it corresponds to the divergence of the damping time at the threshold which is found for bunched beams [67]. In the three cases the PIC simulations reproduce the solutions of the linearized equations to within about one percent, as long as the response exceeds the fluctuations of the reference simulation.

Next, we consider the polytropes with $M_{gas}/M = 0.01$, which is the regime to which the results of Ref. [31] refer. In this case the gain is small (for $\delta \geq 10^{-3}$ it does not exceed 0.06), the response is dominated by mixing, and its behavior for large times is determined by the edge of the support of the DF. For the mixing in the fixed potential Φ_{eq}, Eq. (33) applies with $\hat{F}_1 = \mathcal{F}_{0,eq}\, s/2$, which according to Eq. (45) corresponds to $g(J_{max}) = \alpha\, \Omega_{min}^k B(J_{max})/2$ in Eq. (32), and one obtains

$$|\chi_1(t)| \simeq \frac{C}{t^{k+1}}, \qquad C := \frac{\Gamma(k+1)\, \alpha\, \Omega_{min}^k B(J_{max})}{2\, |\Omega'(J_{max})|^{k+1} \int_0^{J_{max}} \mathcal{F}_{0,eq}(J) B(J)\, dJ}, \qquad t \to \infty. \tag{78}$$

We have verified this expression by solving Eq. (36) without its third term: at $t = 4000$ the envelope of $|\chi_1|$ exceeds the right-hand side of Eq. (78) by 4 to 6 percent for the four values of k, and the difference decreases approximately like $1/t$, as expected for the next term of the asymptotic expansion.

The self-gravity of the perturbation modifies this behavior. To see how, we expand $\delta\mathcal{F}_0$ and $\delta\Phi(t, r(Q,J))$ in Fourier series with respect to Q, with coefficients $\delta\hat{F}_n(t,J)$ and $\phi_n(t,J)$, respectively. Equation (36) then yields $\partial_t \delta\hat{F}_n + in\Omega\,\delta\hat{F}_n = in\,\mathcal{F}'_{0,eq}\phi_n$, whose solution is

$$\delta\hat{F}_n(t,J) = e^{-in\Omega(J)t}\left[\delta\hat{F}_n(0,J) + in\,\mathcal{F}'_{0,eq}(J)\,\hat{\Phi}_n(t,J)\right], \qquad \hat{\Phi}_n(t,J) := \int_0^t \phi_n(t',J)e^{in\Omega(J)t'}\,dt'. \tag{79}$$

Hence, the perturbation evolves like in a fixed potential, but with an effective initial datum which accumulates the action of the perturbed potential on each orbit. The important point is that $\mathcal{F}'_{0,eq}$ vanishes like $(J_{max}-J)^{k-1}$ at the edge of the support, that is, one power more slowly than the initial perturbation. Suppose that $k>1$ and that $\hat{\Phi}_1(t,J_{max})$ converges to a limit $\hat{\Phi}_\infty \neq 0$ for $t\to\infty$. Then, the argument which leads to Eq. (33), applied to both terms on the right-hand side of Eq. (79), gives

$$\chi_1(t) \simeq \chi_1^{mix}(t)\left[1 + 2|\Omega'(J_{max})|\,\hat{\Phi}_\infty\,t\right], \tag{80}$$

where χ_1^{mix} denotes the asymptotic expression for the mixing in the fixed potential, whose magnitude is given by Eq. (78). Therefore, for times which are large compared to $t_\times := (2|\Omega'(J_{max})||\hat{\Phi}_\infty|)^{-1}$ the response decays like t^{-k} instead of $t^{-(k+1)}$: the self-gravity of the perturbation reduces the exponent of the algebraic tail by one. In the derivation of Eq. (80) we have treated $\hat{\Phi}_1(t,J)$ as a regular function of J which is independent of t at late times. This neglects the potential which is generated by the tail itself; hence, the amplitude of the second term is only correct to leading order in M_{gas}/M.

The exponent $-k$ is not new as a decay rate. For perturbations which are proportional to $\mathcal{F}'_{0,eq}$, like the second term in Eq. (79), the estimates of Ref. [23] give the decay rate $t^{-\min\{2,k\}}$ for the time derivative of the potential, and in the Laplace representation of Refs. [51, 55] the exponent results from the singularity of the response function at the edge of the band, which contains $\mathcal{F}'_{0,eq}$. What the present analysis adds is that the self-gravity of the perturbation converts the tail $t^{-(k+1)}$ of a perturbation which is proportional to the DF of the steady state into t^{-k}, together with the time $t_\times$ at which this occurs. We are not aware of quantitative results on the decay of the solutions of the full linearized system for steady states of this kind; see the remarks in Ref. [34] and, for steady states with a distribution of angular momenta around a point mass, Ref. [32]. [(Erik) The derivation of Eq. (80) is formal and should be checked by Olivier. The Laplace representation would also give the amplitude of the tail beyond the leading order in M_{gas}/M; this has not been done.]

The solutions of the full linearized equations for $M_{gas}/M = 0.01$ are shown in Fig. 6(b). For $k=1.5$ and $k=2$ the function $\hat{\Phi}_1(t,J_{max})$, which we compute from the potential along the outermost rows of the lattice, converges for $t\geq 2000$, and the resulting times $t_\times = 3.5\times10^4$ and 6.0×10^4 exceed the time span of the solutions. Accordingly, the tail follows the law of the mixing in a fixed potential: the exponents which are obtained from a fit of the envelope in the interval $2000\leq t\leq 4000$ are -2.58 and -3.10, both with and without self-gravity, where the difference with respect to $-(k+1)$ is due to the correction of the order $1/t$. For $k\leq 1$ the function $\hat{\Phi}_1(t,J_{max})$ does not converge, and the second term in Eq. (80), evaluated with $\hat{\Phi}_1(t,J_{max})$ in place of $\hat{\Phi}_\infty$, grows to 0.75 at $t=8000$ for $k=0.75$. It accounts for the difference with respect to the mixing in a fixed potential: the envelope with self-gravity exceeds the one without it by a factor which grows from 1.06 at $t=2000$ to 1.35 at $t=8000$ for $k=0.75$, and to 1.19 for $k=1$. This slow growth is the only trace of the oscillating mode which we find for these steady states. Over the time span considered here the perturbation decays for all the values of k, approximately like $t^{-(k+1)}$, and the dichotomy between $k\leq 1$ and $k>1$ cannot be observed. The PIC simulations agree with the solutions of the linearized equations until the response has decreased by about three orders of magnitude (see section V D), such that they confirm the decay and the onset of the tail, whereas the exponents rely on the linearized equations.

The change of the exponent is observed for larger masses. For the polytrope with $k=2$ and $M_{gas}/M=1$, for which $\lambda_{edge}=0.820$ and $t_\times = 65$, the response starts with a damped oscillation, for which a fit in the interval $100\leq t\leq 400$ gives $\omega = 0.1668\pm0.0008$ and $\gamma = (6.4\pm0.8)\times10^{-3}$ for the linearized equations and $\omega = 0.1671\pm0.0005$ and $\gamma = (6.4\pm0.9)\times10^{-3}$ for the PIC simulation with $\varepsilon=0.1$. For $t>1500$ it follows an algebraic tail which oscillates with the frequency Ω_{min}, see Fig. 6(c). The exponent of its envelope is -2.07 in the interval $2000\leq t\leq 4000$ and -2.04 in $4000\leq t\leq 8000$, that is, the exponent $-k=-2$ of Eq. (80) and not the exponent -3 of the mixing in a fixed potential, whose amplitude (78) the tail exceeds by a factor of 400 at $t=2000$ and of 1500 at $t=8000$. As expected for $M_{gas}/M=1$, Eq. (80) does not reproduce the amplitude of the tail, which is about 13 times larger than the one which results from the computed value $|\hat{\Phi}_\infty| = 0.047$. The same change of the exponent is found for the polytrope with $k=1.5$ and $M_{gas}/M=0.6$, for which $\lambda_{edge}=0.759$: the exponent of the envelope is -1.54 in the interval $2000\leq t\leq 4000$ and -1.52 in $4000\leq t\leq 8000$, compared to -2.5 for the mixing in a fixed potential. [(Erik)

The PIC simulation for this case has $N = 10^4$ and $\varepsilon = 0.1$. It is to be replaced by the simulations with $N = 1.024 \times 10^5$ and $\varepsilon = 0.03$ and 0.1 which have been run on the cluster and whose analysis is pending.]

D. Oscillating modes above the threshold

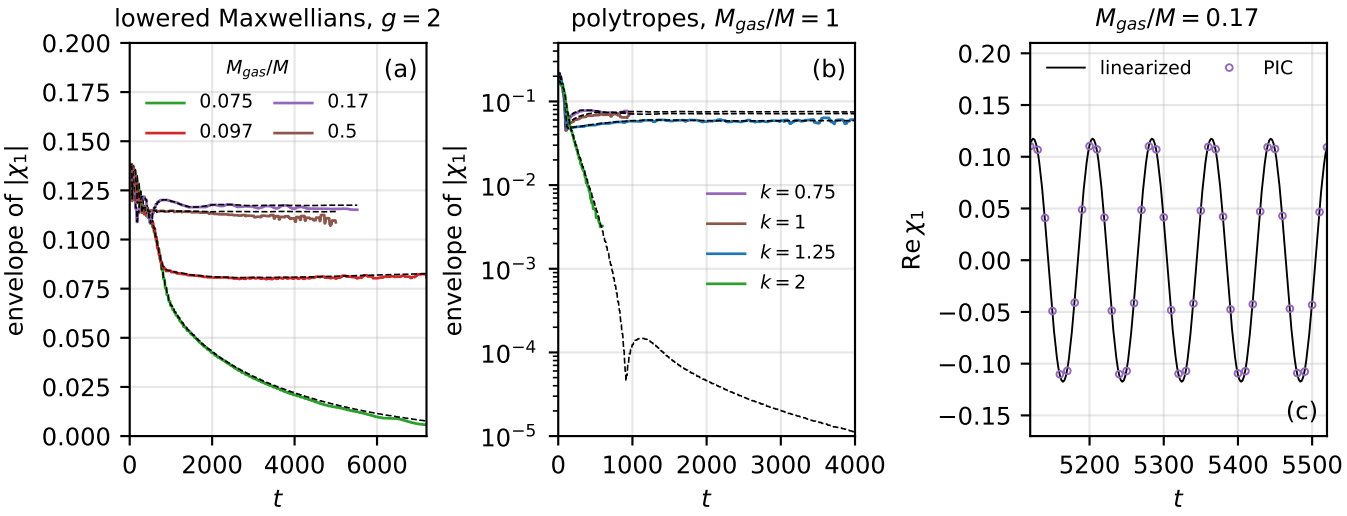


FIG. 7: Response of the gas for steady states above the threshold mass. (a) Envelope of $|\chi_1(t)|$ (largest value in a moving window of width 100) for the lowered Maxwellians with $g = 2$ and $J_{max} = 0.138$, obtained from the PIC simulations with $\varepsilon = 0.075$, 0.01, 0.1 and 0.1 in the order of increasing mass (solid lines) and from the linearized equations (dashed lines). The case $M_{gas}/M = 0.075$, which lies below the threshold, is included for comparison. (b) The same for the polytropes with $M_{gas}/M = 1$, with a window of width 50. The PIC simulations have $\varepsilon = 0.1$ and are shown for $t \le 1000$ ($t \le 600$ for $k = 2$), except for $k = 1.25$, where $\varepsilon = 0.01$ and $N = 1.024 \times 10^5$. (c) Real part of χ_1 at the end of the simulation for the lowered Maxwellian with $M_{gas}/M = 0.17$.

When $\lambda_{edge} > 1$ the response does not decay, see Figs. 7(a) and (b) for the lowered Maxwellians with $M_{gas}/M = 0.097$, 0.17 and 0.5 and for the polytropes with $k = 0.75$, 1 and 1.25 and $M_{gas}/M = 1$. After a transient, during which the part of the perturbation that does not belong to the mode undergoes mixing, the envelope of $|\chi_1|$ settles down to a constant value, which lies between 59 and 85 percent of $|\chi_1(0)|$ for the lowered Maxwellians and between 30 and 34 percent for the polytropes. The oscillation persists until the end of the linearized solutions, which comprise between 70 and 100 periods $2\pi/\omega_d$, and the PIC simulations follow them in amplitude as well as in phase, see Fig. 7(c). In these cases the mixing of the perturbation is not completed.

		Ω_{min}	δ_d	ω_d from Eq. (44)	linearized equations	PIC
lowered Maxwellian, $g = 2$	$M_{gas}/M = 0.097$	0.072712	0.0124	0.072496	0.072495	0.072480
	$M_{gas}/M = 0.17$	0.082745	0.187	0.078478	0.078477	0.078471
	$M_{gas}/M = 0.5$	0.134193	0.585	0.104856	0.104875	0.104823
polytrope, $M_{gas}/M = 1$	$k = 0.75$	0.151589	0.0750	0.141686	0.141690	0.14157
	$k = 1$	0.153987	0.0439	0.147851	0.147835	0.14776
	$k = 1.25$	0.156012	0.0187	0.153259	0.153252	0.153212
	$k = 1.5$	0.157741	0.00147	0.157515	0.157514	–

TABLE IV: Frequency of the oscillating mode obtained from three different methods: the equation $\lambda(\omega_d) = 1$, a fit of the solution of the linearized equations, and a fit of the function $\chi_1(t)$ of the PIC simulations, both with the method of section IV F. For the lowered Maxwellians the fits refer to the second half of the time span of the simulations, which extend to $t = 7200$, 5520 and 5000 and have the amplitudes $\varepsilon = 0.01$, 0.1 and 0.1, respectively. For the polytropes, the linearized solutions are fitted in the interval $1000 \le t \le 4000$ ($8000 \le t \le 14000$ for $k = 1.5$) and the PIC simulations in $300 \le t \le 1000$ for $k = 0.75$ and $k = 1$ ($N = 10^4$, $\varepsilon = 0.1$) and in $500 \le t \le 3900$ for $k = 1.25$ ($N = 1.024 \times 10^5$, $\varepsilon = 0.01$).

Table IV compares the frequencies of the oscillation. Those of the linearized solutions agree with the roots of the equation $\lambda(\omega_d) = 1$ to within 2×10^{-5}, that is, to within 2×10^{-4} in relative terms, and the damping rates which

result from the same fits are smaller than 5×10^{-6} in magnitude. The frequencies of the PIC simulations lie between 7×10^{-6} and 5×10^{-5} below ω_d, except for the two simulations of the polytropes with $\varepsilon = 0.1$ and $N = 10^4$, where the difference is about 10^{-4}, in line with the decrease of the frequency with the amplitude found in section V D. In all the cases the frequency lies below Ω_{min}, and the distance between both is resolved by the three methods. An agreement of this kind between the linear theory and N-body simulations was obtained for the modes of the one-dimensional slab in Refs. [39, 65]. Together with the results of the previous subsection this confirms the criterion of section III C: for the lowered Maxwellians with $g = 2$ and $J_{max} = 0.138$ the response is damped for $M_{gas}/M = 0.075$ and oscillates without damping for $M_{gas}/M = 0.097$, in accordance with the threshold 0.0906, and for the polytropes with $M_{gas}/M = 1$ it oscillates for $k \leq 1.5$ and is damped for $k = 2$. For the polytrope with $k = 1.5$, which lies only slightly above its threshold mass 0.920 ($\delta_d = 1.5 \times 10^{-3}$), a time of the order of $(\Omega_{min} - \omega_d)^{-1} = 4400$ is needed for the mode to separate from the contribution of the orbits near the edge (see section III D): the envelope of the linearized solution decreases until $t \approx 4000$ and then remains between 12 and 13 percent of $|\chi_1(0)|$ up to $t = 1.4 \times 10^4$, with a frequency of 0.157514, in agreement with $\omega_d = 0.157515$. A PIC simulation of this mode in the linear regime would require a much smaller amplitude than those used here, as we show next. [(Erik) The PIC values for $k = 0.75$ and $k = 1$ are to be replaced by those of the simulations with $N = 1.024 \times 10^5$ and $\varepsilon = 0.03$ which have been run on the cluster and whose analysis is pending.]

E. Finite amplitude: trapping of the orbits near the edge

In section V D we found that for the polytrope with $k = 1.25$ and $M_{gas}/M = 1$ the amplitude of the response falls below the one predicted by the linearized equations by an amount which depends on ε and which is not a numerical artifact. In this subsection we show that this effect is caused by the gas particles near the edge of the steady state, which are trapped by the oscillating mode when its amplitude exceeds a critical value. That the amplitude of an oscillating mode which lies close to the band is limited was observed by Weinberg [39] in N-body simulations of the one-dimensional slab, who attributed this limit to the nonlinear broadening of the frequency of the mode, and Louis [65] pointed out that the trapping of resonant particles leads to oscillations of the amplitude. The pendulum model which follows turns these observations into an explicit condition.

Consider the motion of a gas particle under the influence of an oscillating mode with frequency ω_d. In terms of the action-angle variables of the steady state, this motion is governed by the Hamiltonian $H = E(J) + \delta\Phi(t, r(Q, J))$, see section III B, and the expansion of section III C gives

$$\delta\Phi(t, r(Q, J)) = \varepsilon \sum_{n \in \mathbb{Z}} \phi_n(J) e^{i(nQ - \omega_d t)} + c.c., \tag{81}$$

where $\phi(r)$ refers to the mode which is excited by the perturbation (61) with $\varepsilon = 1$ in the linear regime. For an orbit whose frequency $\Omega(J)$ is close to ω_d, the term with $n = 1$ in Eq. (81) and its complex conjugate vary slowly along the unperturbed motion $Q = Q_0 + \Omega(J)t$, whereas all the other terms oscillate with frequencies which are at least of the order of ω_d and average out. Keeping only the former, writing $\phi_1(J) = |\phi_1(J)|e^{i\beta}$ and introducing the angle

$$\vartheta := Q - \omega_d t + \beta, \tag{82}$$

which together with J constitutes a set of canonical variables, one obtains the time-independent Hamiltonian

$$K(\vartheta, J) = E(J) - \omega_d J + 2\varepsilon |\phi_1(J)| \cos\vartheta. \tag{83}$$

Let J_r be the action variable of the resonant orbit, that is, $\Omega(J_r) = \omega_d$. Since $\omega_d < \Omega_{min}$ and Ω is a decreasing function, this orbit lies outside the support of the steady state, at the distance

$$d := J_r - J_{max} \approx \frac{\Omega_{min} - \omega_d}{|\Omega'(J_{max})|} \tag{84}$$

from its edge. Expanding $E(J) - \omega_d J$ about $J = J_r$ to second order and neglecting the variation of ϕ_1 and Ω' between J_{max} and J_r, Eq. (83) yields, up to an additive constant,

$$K(\vartheta, J) \approx -\frac{1}{2}|\Omega'|(J - J_r)^2 + 2\varepsilon|\phi_1| \cos\vartheta, \tag{85}$$

which is the Hamiltonian of a pendulum. Its level sets are separated into two classes by the separatrix

$$J = J_r \pm \Delta J_s \left|\cos\frac{\vartheta}{2}\right|, \qquad \Delta J_s := \frac{2\omega_b}{|\Omega'|}, \qquad \omega_b := \sqrt{2\varepsilon|\phi_1||\Omega'|}. \tag{86}$$

A particle which lies inside the separatrix is trapped by the mode: its angle ϑ oscillates about zero, and its action variable oscillates about J_r with an amplitude of up to ΔJ_s and with a frequency which is equal to the bounce frequency ω_b close to the center $(\vartheta, J) = (0, J_r)$ and decreases towards the separatrix. Outside the separatrix ϑ varies monotonically. The same pendulum describes the particles which are trapped by a wave in a plasma, where it determines the nonlinear evolution of Landau damping [68], and the orbits near a resonance in a galaxy [1].

Initially, all the gas particles have $J \leq J_{max} < J_r$. Therefore, the effect of the mode on them depends on the ratio

$$s := \frac{\Delta J_s}{d} = \frac{2\omega_b}{\Omega_{min} - \omega_d} \tag{87}$$

between the half-width of the separatrix and the distance of the resonance from the edge. If $s < 1$, the separatrix does not reach the support of the steady state, and none of the gas particles is trapped. Their action variables oscillate, and it follows from Eq. (85) that the largest value which is attained by a particle with $J = J_{max}$ at $t = 0$ is $J_+ = J_{max} + d\,(1 - \sqrt{1 - s^2})$. If $s > 1$, the particles close to the edge whose angle ϑ lies in a suitable interval about zero are trapped, and those which start next to the separatrix are carried to its top, such that $J_+ = J_r + \Delta J_s = J_{max} + d\,(1 + s)$. Since ω_b is proportional to $\sqrt{\varepsilon}$, the condition $s > 1$ is equivalent to

$$\varepsilon > \varepsilon_c := \frac{(\Omega_{min} - \omega_d)^2}{8|\phi_1||\Omega'|}. \tag{88}$$

Hence, the closer the frequency of the mode lies to the edge of the band, the smaller is the amplitude which is required for trapping to occur.

In order to evaluate these expressions we determine ϕ_1 from the solution of the linearized equations, by means of a least-squares fit of the first harmonic $(1/N_Q)\sum_b \delta\Phi(t, r_{ab})e^{-iQ_b}$ of the perturbation of the potential along the outermost row $a = N_J$ of the lattice. Since the radius is an even function of Q, this quantity is real, and according to Eq. (81) it is equal to $2\,\mathrm{Re}[\phi_1(J)e^{-i\omega_d t}]$ once the transient has decayed. The results are listed in Table V; they change by less than two percent when the fit is repeated in a subsequent interval, except for $k = 1.5$, where the change is seven percent.

		$\Omega_{min} - \omega_d$	$\lvert\Omega'(J_{max})\rvert$	$\lvert\phi_1\rvert$	$\omega_b/\sqrt{\varepsilon}$	ε_c	J_r/J_{max}
lowered Maxwellian, $g = 2$	$M_{gas}/M = 0.097$	2.17×10^{-4}	0.0976	1.60×10^{-5}	1.77×10^{-3}	0.0037	1.016
	$M_{gas}/M = 0.17$	4.27×10^{-3}	0.1193	4.16×10^{-5}	3.15×10^{-3}	0.46	1.26
	$M_{gas}/M = 0.5$	2.93×10^{-2}	0.2306	1.31×10^{-4}	7.78×10^{-3}	3.6	1.92
polytrope, $M_{gas}/M = 1$	$k = 0.75$	9.90×10^{-3}	0.1422	7.51×10^{-4}	1.46×10^{-2}	0.115	1.100
	$k = 1$	6.14×10^{-3}	0.1475	5.76×10^{-4}	1.30×10^{-2}	0.056	1.060
	$k = 1.25$	2.75×10^{-3}	0.1520	4.06×10^{-4}	1.11×10^{-2}	0.015	1.026
	$k = 1.5$	2.3×10^{-4}	0.1557	1.45×10^{-4}	6.7×10^{-3}	3×10^{-4}	1.002

TABLE V: Parameters of the pendulum (85) which describes the motion of the gas particles near the edge of the steady state in the field of the oscillating mode: distance of the frequency of the mode from the edge of the band, derivative of the frequency of the orbits and magnitude of the first harmonic of the potential of the mode at the edge (for $\varepsilon = 1$), bounce frequency, critical amplitude (88) and action variable of the resonant orbit.

The critical amplitude varies over four orders of magnitude. For the lowered Maxwellian with $M_{gas}/M = 0.5$ it is larger than one, such that trapping does not occur for the perturbation (61), whereas for the modes which lie close to the edge it is very small: $\varepsilon_c = 0.0037$ for the lowered Maxwellian with $M_{gas}/M = 0.097$, and $\varepsilon_c = 0.015$ and 3×10^{-4} for the polytropes with $k = 1.25$ and $k = 1.5$. In particular, of the three amplitudes $\varepsilon = 0.01$, 0.03 and 0.1 which were compared in section V D for $k = 1.25$, the first one lies below the critical amplitude ($s = 0.81$) and the other two above it ($s = 1.40$ and 2.55).

These estimates are compared with the PIC simulations in Table VI and Fig. 8. Figure 8(c) shows the computational particles near the edge for the polytrope with $k = 1.25$ and $\varepsilon = 0.1$ at half a bounce period, $t = 900$: those which were located next to the edge with angles close to $\vartheta = 0$ have moved around the center of the island and reach $J = 0.76$, while the top of the separatrix lies at $J_r + \Delta J_s = 0.764$. The largest value of the action variable which is attained in the simulations agrees with the estimate J_+, see Fig. 8(d): for the simulations with $s > 1$ which extend over at least two units of $1/\omega_b$ the excursion $J_+ - J_{max}$ differs from the estimate by at most 12 percent, and for $k = 1.25$ it grows by a factor of five between $\varepsilon = 0.01$ and $\varepsilon = 0.03$, which reflects the onset of the trapping at $\varepsilon_c = 0.015$. We stress that the estimate does not contain any adjustable parameter.

The trapping is accompanied by a loss of amplitude of the mode. In the simulations with $s < 1$ the ratio (76) between the amplitudes of the responses of the PIC code and of the linearized equations remains close to one (its

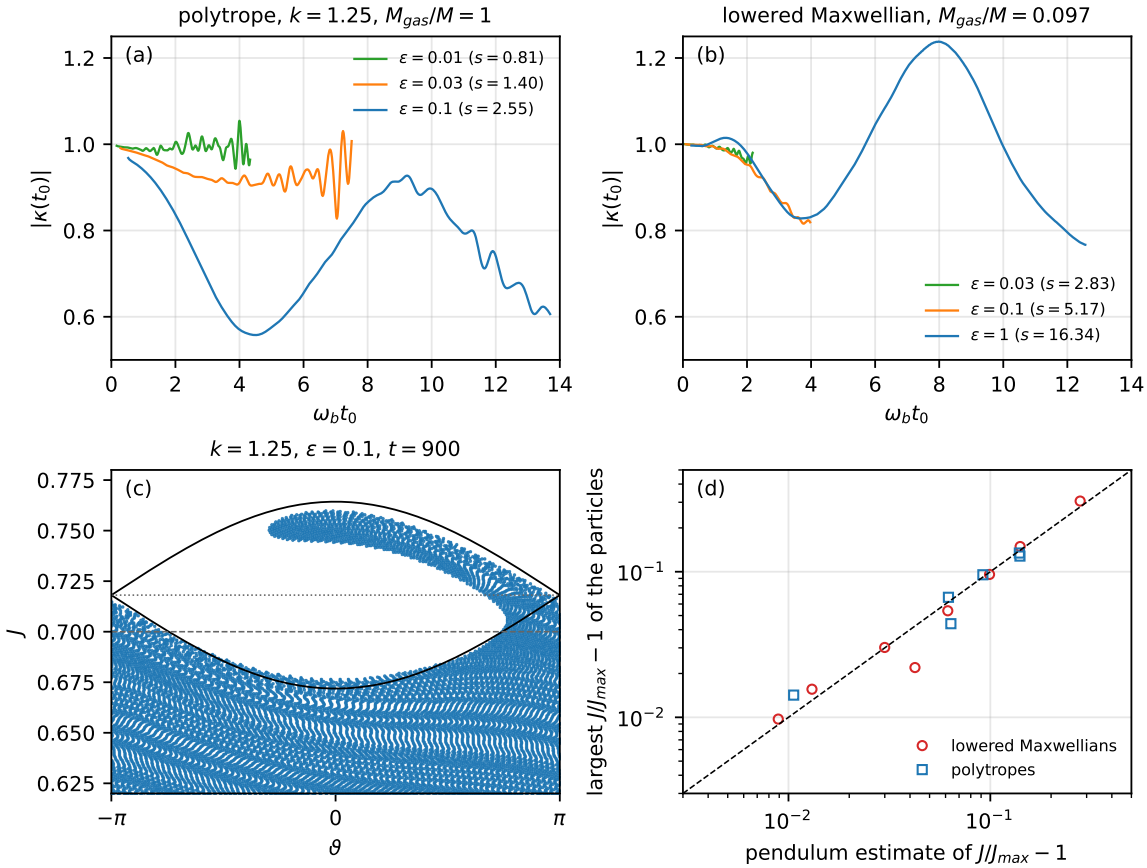


FIG. 8: Effect of the amplitude of the perturbation on the oscillating mode. (a) Ratio $|\kappa(t_0)|$ between the amplitudes of the response obtained from the PIC code and from the linearized equations, see Eq. (76), as a function of time in units of the inverse bounce frequency, for the polytrope with $k = 1.25$ and $M_{gas}/M = 1$ and three amplitudes ($N = 1.024 \times 10^5$). (b) The same for the lowered Maxwellian with $g = 2$ and $M_{gas}/M = 0.097$ ($N = 10^4$). (c) The computational particles near the edge of the steady state for the simulation with $\varepsilon = 0.1$ of panel (a) at $t = 900$, which corresponds to half a bounce period, in terms of the angle (82) and the action variable. The solid lines are the separatrix (86), the dashed line indicates J_{max} and the dotted one J_r. (d) Largest value of the action variable attained by the computational particles in the simulations of Table VI, compared to the value J_+ predicted by the pendulum (85).

smallest value is 0.96, see Table VI), whereas for $s > 1$ the amplitude falls below the linear one on the time scale $1/\omega_b$ and passes through a minimum at $\omega_b t_0 \approx 4$, see Figs. 8(a) and (b). For the polytrope with $k = 1.25$ this minimum is 0.56 for $\varepsilon = 0.1$ and 0.89 for $\varepsilon = 0.03$. For the lowered Maxwellian with $M_{gas}/M = 0.097$ the curves for $\varepsilon = 0.03$, 0.1 and 1 coincide with each other to within four percent when they are represented as functions of $\omega_b t_0$, and in the longest simulation ($\varepsilon = 1$) the minimum $|\kappa| = 0.83$ is followed by a maximum of 1.24 at $\omega_b t_0 = 8.0$, such that the amplitude of the mode oscillates with a period which is about 1.4 times the bounce period $2\pi/\omega_b$. Such oscillations are characteristic for a wave which interacts with trapped particles [68]; for the one-dimensional slab they were anticipated in Ref. [65]. For $\varepsilon = 0.1$ and $M_{gas}/M = 0.097$ we have verified that $|\kappa|$ does not change when the number of particles is multiplied by four or the time step is halved.

The reason for the loss of amplitude is the following. According to Eq. (38), in the linearized theory the DF of the mode is proportional to $\mathcal{F}'_{0,eq}(J)\,\phi_1(J)/[\Omega(J) - \omega_d]$, which is largest for the orbits near the edge, where $\Omega(J) - \omega_d$ is smallest. This expression presupposes that the particles follow the unperturbed orbits. For the trapped particles this is not the case: their contribution to the density of the mode ceases to grow in phase with the potential and oscillates with the bounce period instead. Although these particles carry a very small fraction of the mass of the gas (in the simulation with $k = 1.25$ and $\varepsilon = 0.1$, the particles which reach $J > J_{max} + 0.003$ amount to $1.5 \times 10^{-4} M_{gas}$), their weight in the mode is large due to the small denominator. Consistently, at comparable values of s the loss is larger for the polytrope with $k = 1.25$ than for the lowered Maxwellian with $g = 2$, since in the former case the derivative of the DF vanishes more slowly at the edge. This is also in line with the analysis by Louis and Gerhard [69] of a radial oscillation of finite amplitude of the isochrone sphere, in which the orbits that are trapped around the stable resonant

		ε	s	J_+ (PIC)	J_+ (pendulum)	min $\lvert\kappa\rvert$	$\omega_b t_0$
lowered Maxwellian, $g=2$	$M_{gas}/M=0.097$	0.003	0.89	0.1393	0.1392	0.98	–
		0.01	1.63	0.1410	0.1438	0.98	1.2*
		0.03	2.83	0.1454	0.1465	0.96	2.1*
		0.1	5.17	0.1512	0.1517	0.82	3.8*
		1	16.3	0.1802	0.1765	0.83	3.7
	$M_{gas}/M=0.17$	0.1	0.47	0.1422	0.1421	0.99	–
	$M_{gas}/M=0.5$	0.1	0.17	0.1402	0.1398	0.99	–
		1	0.53	0.1586	0.1574	1.00	–
polytrope, $M_{gas}/M=1$	$k=1.25$	0.01	0.81	0.7099	0.7074	0.98	–
		0.03	1.40	0.7467	0.7434	0.89	5.4
		0.1	2.55	0.7667	0.7643	0.56	4.5
		0.3	4.42	0.7897	0.7981	0.31	4.1
	$k=1$	0.1	1.34	0.7941	0.7975	0.89	3.6*
	$k=0.75$	0.1	0.93	0.7308	0.7446	0.96	–

TABLE VI: Simulations of steady states with an oscillating mode for different amplitudes of the perturbation: ratio (87), largest value of the action variable attained by the computational particles and its estimate J_+ from the pendulum (85), and smallest value of the ratio $|\kappa|$ between the amplitudes of the response of the PIC code and of the linearized equations. For $s>1$ the latter refers to the interval $\omega_b t_0 \le 6$, and the last column gives the time at which it is attained; an asterisk indicates that this occurs at the end of the time span which is analyzed, such that $|\kappa|$ may decrease further. For $k=1.25$ and $\varepsilon=0.01$ the smallest value for $t_0 \le 2000$ is given. The simulations have $N=10^4$ and are analyzed over their whole duration, except those of the polytropes with $k=1.25$ and $\varepsilon \le 0.1$, which have $N=1.024\times10^5$, and the remaining ones of the polytropes, which are only analyzed for $t \le 1000$.

orbits oscillate against the perturbation and damp it, such that a self-consistent oscillation requires a DF which is depleted at the resonances. However, we do not have a quantitative theory for the depth of the minimum.

Finite amplitude effects are also present below the threshold, where the resonant orbit lies inside the support of the steady state. For the lowered Maxwellian with $M_{gas}/M=0.075$, whose response in the linear regime is damped with the rate $\gamma=3.4\times10^{-4}$, we have performed a simulation with $\varepsilon=0.84$ up to $t=17200$. In this case the decay of the envelope of $|\chi_1|$ stops at $t\approx2700$, when it has decreased to 22 percent of its initial value. Afterwards the envelope grows to 52 percent at $t\approx6500$, decreases to 18 percent at $t\approx11500$ and grows again. The period of this oscillation, about 8900, is comparable to the bounce period $2\pi/\omega_b=7700$ which results from Eq. (86) with the mean amplitude of the potential in the interval $4000\le t\le17200$; the same values are obtained with four times as many particles and with half the time step up to $t=5000$. This is the saturation of Landau damping by the trapping of the resonant particles which was predicted by O'Neil [68] for a plasma wave: the damping stops once the bounce frequency exceeds the damping rate.

We conclude that the linear regime of an oscillating mode is restricted to amplitudes $\varepsilon \ll \varepsilon_c$. For the modes which emerge at the threshold this is a severe restriction, since ε_c vanishes quadratically with the distance $\Omega_{min}-\omega_d$ of the mode from the band. In this sense, Eq. (88) gives an explicit expression for the amplitude at which the limitation observed in Ref. [39] sets in. The trapping is controlled by the ratio between the bounce frequency and a difference of orbital frequencies. In section VII we find that a ratio of the same kind, between the bounce frequency in the potential well of the gas and the spread of the orbital frequencies, determines the evolution of initial data far from equilibrium. [(Erik) The simulations of this subsection for $k=0.75$ and $k=1$ have $N=10^4$ and $\varepsilon=0.1$, and their analysis is restricted to $t\le1000$. They are to be replaced by the simulations with $N=1.024\times10^5$ and $\varepsilon=0.03$ and 0.1 which have been run on the cluster and whose analysis is pending; according to Table V, $\varepsilon=0.03$ lies below the critical amplitude in both cases.]

VII. INITIAL DATA FAR FROM EQUILIBRIUM

[(Erik) To be written: the blob concentrated about one orbital phase (runs B0–B5 of the demo document and the new runs), the transition from mixing to a self-bound core as a function of μ, its dependence on the widths of the initial DF, convergence, and the final state.]

VIII. CONCLUSIONS

[(Erik) To be written.]

Acknowledgments

[(Erik) To be written.]

Appendix A: Numerical computation of the action-angle variables

In this appendix we describe the numerical evaluation of the action-angle variables defined in Eqs. (16,17,18) for a potential of the form $\Phi = \Phi_{ext} + \Phi_{gas}$, in which Φ_{gas} is only known on a radial grid. This is required for the construction of the self-consistent steady states, for the generation of the initial data and for the computation of the observables $h_n(t)$.

The values of Φ_{gas} on the grid are interpolated with a natural cubic spline, and beyond the last grid point we set $\Phi_{gas} = -GM_{gas}/r$. Given a point (r, p_r, L) we compute its energy E from Eq. (15), locate the minimum $r_{min}(L)$ of the effective potential V_L, and determine the turning points $r_1 < r_{min}(L) < r_2$ by solving the equation $V_L(r) = E$ with the bisection method on either side of the minimum. The integrand in Eq. (16) behaves like a square root at the turning points and the one in Eq. (17) like an inverse square root, which spoils the convergence of standard quadrature rules. We avoid this problem by means of the substitution

$$r = r_m + r_a \sin\theta, \qquad r_m := \frac{r_1 + r_2}{2}, \quad r_a := \frac{r_2 - r_1}{2}, \qquad -\frac{\pi}{2} \le \theta \le \frac{\pi}{2}. \tag{A1}$$

Since for a non-circular orbit the zeros of the function $E - V_L$ at the turning points are simple, one has $E - V_L(r) = r_a^2 \cos^2\theta\, u(\theta)$ with a smooth and positive function u, and it follows that

$$A(E,L) = 2r_a^2 \int_{-\pi/2}^{\pi/2} \cos^2\theta \sqrt{2u(\theta)}\, d\theta, \qquad T(E,L) = 2 \int_{-\pi/2}^{\pi/2} \frac{d\theta}{\sqrt{2u(\theta)}}. \tag{A2}$$

Both integrands are smooth, and we evaluate the integrals with a Gauss-Legendre rule with 64 nodes. The angle variable is obtained in the same way from

$$Q = \frac{2\pi}{T(E,L)} \int_{-\pi/2}^{\theta(r)} \frac{d\theta'}{\sqrt{2u(\theta')}}, \qquad \theta(r) := \arcsin\left(\frac{r - r_m}{r_a}\right), \tag{A3}$$

when $p_r \ge 0$, and from 2π minus the right-hand side of Eq. (A3) when $p_r < 0$. For the isochrone potential without self-gravity, for which the exact expressions (22,23) are available, this procedure yields the action variable with an error of the order of 10^{-14} and the period with an error of the order of 10^{-11}. Without the substitution (A1) the corresponding errors are 2×10^{-6} and 9×10^{-3}, and they only decrease like the third and the first inverse power of the number of nodes, respectively.

For the inverse map $(Q, J) \mapsto (r, p_r)$ at fixed L we tabulate the function $J = A(E,L)/2\pi$ for 4000 values of the energy between $E_{min}(L)$ and a value slightly above the cutoff energy, and obtain $E(J,L)$ by interpolation. Next, we determine the turning points and find, by bisection in θ, the point of the orbit at which the time elapsed since the passage through the pericenter is equal to $QT(E,L)/2\pi$ when $Q \le \pi$ and to $(2\pi - Q)T(E,L)/2\pi$ otherwise. This gives the radius r, and the momentum is $p_r = \pm\sqrt{2(E - V_L(r))}$, with the positive sign when $Q \le \pi$. Applying the direct map to the result reproduces the points of the lattice (58) with errors which are smaller than 10^{-8} in J and 5×10^{-7} in Q. The frequency is obtained from $\Omega = 2\pi/T(E,L)$.

[1] J. Binney and S. Tremaine. *Galactic Dynamics*. Princeton University Press, Princeton, second edition, 2008.
[2] G. Rein. Collisionless kinetic equations from astrophysics – the Vlasov-Poisson system. In *Handbook of Differential Equations: Evolutionary Equations*, volume 3, pages 383–476. Elsevier, 2007.
[3] H. Andréasson. The Einstein-Vlasov System/Kinetic Theory. *Living Rev. Relativ.*, 14:4, 2011.
[4] J. R. Oppenheimer and H. Snyder. On Continued Gravitational Contraction. *Phys. Rev.*, 56:455–459, 1939.

[5] D. Christodoulou. Violation of cosmic censorship in the gravitational collapse of a dust cloud. *Commun. Math. Phys.*, 93:171–195, 1984.

[6] N. Ortiz and O. Sarbach. Conformal diagrams for the gravitational collapse of a spherical dust cloud. *Class. Quantum Grav.*, 28:235001, 2011.

[7] K. Pfaffelmoser. Global classical solutions of the Vlasov-Poisson system in three dimensions for general initial data. *J. Differ. Equ.*, 95:281–303, 1992.

[8] P. L. Lions and B. Perthame. Propagation of moments and regularity for the 3-dimensional Vlasov-Poisson system. *Invent. Math.*, 105:415–430, 1991.

[9] J. Schaeffer. Global existence of smooth solutions to the Vlasov Poisson system in three dimensions. *Commun. Partial Differ. Equ.*, 16:1313–1335, 1991.

[10] G. Rein and A. D. Rendall. Global existence of solutions of the spherically symmetric Vlasov-Einstein system with small initial data. *Commun. Math. Phys.*, 150:561–583, 1992.

[11] H. Andréasson, M. Kunze, and G. Rein. The formation of black holes in spherically symmetric gravitational collapse. *Math. Ann.*, 350:683–705, 2011.

[12] D. Lynden-Bell. The Stability and Vibrations of a Gas of Stars. *Mon. Not. R. Astron. Soc.*, 124:279–296, 1962.

[13] D. Lynden-Bell. Statistical Mechanics of Violent Relaxation in Stellar Systems. *Mon. Not. R. Astron. Soc.*, 136:101–121, 1967.

[14] P. Rioseco and O. Sarbach. Phase space mixing in an external gravitational central potential. *Class. Quantum Grav.*, 37:195027, 2020.

[15] S. Tremaine, M. Hénon, and D. Lynden-Bell. H-functions and mixing in violent relaxation. *Mon. Not. R. Astron. Soc.*, 219:285–297, 1986.

[16] C. Mitchell. Weak convergence to equilibrium of statistical ensembles in integrable Hamiltonian systems. *J. Math. Phys.*, 60:052702, 2019.

[17] P. Domínguez-Fernández, E. Jiménez-Vázquez, M. Alcubierre, E. Montoya, and D. Núñez. Description of the evolution of inhomogeneities on a dark matter halo with the Vlasov equation. *Gen. Relativ. Gravit.*, 49:123, 2017.

[18] A. L. García-Perciante, M. Guerrero, D. Núñez, and O. Sarbach. Relaxation of a Vlasov gas to an inhomogeneous state due to phase space mixing in an axisymmetric potential: A Newtonian analogy of the Kerr orbital motion. *arXiv:2610.06913*, 2026.

[19] P. Rioseco and O. Sarbach. Phase space mixing in the equatorial plane of a Kerr black hole. *Phys. Rev. D*, 98:124024, 2018.

[20] P. Rioseco and O. Sarbach. Phase Space Mixing of a Vlasov Gas in the Exterior of a Kerr Black Hole. *Commun. Math. Phys.*, 405:105, 2024.

[21] M. Moreno, P. Rioseco, and H. Van Den Bosch. Mixing in anharmonic potential well. *J. Math. Phys.*, 63:071502, 2022.

[22] S. Chaturvedi and J. Luk. Phase mixing for solutions to 1D transport equation in a confining potential. *Kinet. Relat. Models*, 15:403–416, 2022.

[23] M. Hadžić, G. Rein, M. Schrecker, and C. Straub. Quantitative phase mixing for Hamiltonians with trapping. *arXiv:2405.17153*, 2024.

[24] L. D. Landau. On the vibrations of the electronic plasma. *J. Phys. (USSR)*, 10:25–34, 1946.

[25] C. Mouhot and C. Villani. On Landau damping. *Acta Math.*, 207:29–201, 2011.

[26] M. Hadžić, G. Rein, and C. Straub. On the Existence of Linearly Oscillating Galaxies. *Arch. Ration. Mech. Anal.*, 243:611–696, 2022.

[27] M. Kunze. *A Birman-Schwinger Principle in Galactic Dynamics*, volume 77 of *Progress in Mathematical Physics*. Birkhäuser, Cham, 2021.

[28] M. Kunze. A Birman–Schwinger principle in galactic dynamics: ESI, Vienna, 07–11 February 2022. *Class. Quantum Grav.*, 39:244001, 2022.

[29] S. D. Mathur. Existence of oscillation modes in collisionless gravitating systems. *Mon. Not. R. Astron. Soc.*, 243:529–536, 1990.

[30] M. Moreno, P. Rioseco, and H. Van Den Bosch. Spectrum of the linearized Vlasov–Poisson equation around steady states from galactic dynamics. *J. Math. Phys.*, 67:091501, 2026.

[31] M. Hadžić, G. Rein, M. Schrecker, and C. Straub. Damping Versus Oscillations for a Gravitational Vlasov–Poisson System. *Arch. Ration. Mech. Anal.*, 249:45, 2025.

[32] M. Hadžić and M. Schrecker. On quantitative linear gravitational relaxation. *arXiv:2505.14856*, 2025.

[33] T. Ramming and G. Rein. Oscillating solutions of the Vlasov–Poisson system—A numerical investigation. *Physica D*, 365:72–79, 2018.

[34] C. Straub. Numerical experiments on stationary, oscillating, and damped spherical galaxy models. *Physica D*, 470:134351, 2024.

[35] S. Chaturvedi and J. Luk. Linear and Nonlinear Phase Mixing for the Gravitational Vlasov–Poisson System Under an External Kepler Potential. *Arch. Ration. Mech. Anal.*, 250:68, 2026.

[36] Y. Levin, R. Pakter, F. B. Rizzato, T. N. Teles, and F. P. C. Benetti. Nonequilibrium statistical mechanics of systems with long-range interactions. *Phys. Rep.*, 535:1–60, 2014.

[37] M. Hénon. Numerical experiments on the stability of spherical stellar systems. *Astron. Astrophys.*, 24:229–238, 1973.

[38] P. Klinko and B. N. Miller. Angular-momentum-induced phase transition in spherical gravitational systems: N-body simulations. *Phys. Rev. E*, 65:056127, 2002.

[39] M. D. Weinberg. Vertical oscillation of the Galactic disk. *Astrophys. J.*, 373:391–404, 1991.

[40] S. L. Shapiro and S. A. Teukolsky. Relativistic stellar dynamics on the computer. I - Motivation and numerical method. *Astrophys. J.*, 298:34–57, 1985.

[41] G. Rein, A. D. Rendall, and J. Schaeffer. Critical collapse of collisionless matter: A numerical investigation. *Phys. Rev. D*, 58:044007, 1998.

[42] I. Olabarrieta. Critical collapse of collisionless matter in spherical symmetry. Master's thesis, University of British Columbia, 2000. arXiv:gr-qc/0012059.

[43] I. Olabarrieta and M. W. Choptuik. Critical phenomena at the threshold of black hole formation for collisionless matter in spherical symmetry. *Phys. Rev. D*, 65:024007, 2002.

[44] H. Andréasson and G. Rein. A numerical investigation of the stability of steady states and critical phenomena for the spherically symmetric Einstein–Vlasov system. *Class. Quantum Grav.*, 23:3659–3677, 2006.

[45] C. K. Birdsall and A. B. Langdon. *Plasma Physics via Computer Simulation*. CRC Press, Boca Raton, 1991.

[46] R. W. Hockney and J. W. Eastwood. *Computer Simulation Using Particles*. Adam Hilger, Bristol, 1988.

[47] G. Lapenta. Kinetic plasma simulation: particle in cell method. In *XII Carolus Magnus Summer School on Plasma and Fusion Energy Physics*. 2015.

[48] M. Shalaby, A. E. Broderick, P. Chang, C. Pfrommer, A. Lamberts, and E. Puchwein. SHARP: A Spatially Higher-order, Relativistic Particle-in-cell Code. *Astrophys. J.*, 841:52, 2017.

[49] G. Rein and T. Rodewis. Convergence of a particle-in-cell scheme for the spherically symmetric Vlasov-Einstein system. *Indiana Univ. Math. J.*, 52:821–861, 2003.

[50] F. Leeuwin, F. Combes, and J. Binney. N-body simulations with perturbation particles - I. Method and tests. *Mon. Not. R. Astron. Soc.*, 262:1013–1022, 1993.

[51] J. Barré, A. Olivetti, and Y. Y. Yamaguchi. Algebraic damping in the one-dimensional Vlasov equation. *J. Phys. A: Math. Theor.*, 44:405502, 2011.

[52] M. Hénon. L'amas isochrone: I. *Ann. Astrophys.*, 22:126, 1959.

[53] J. Binney. Hénon's isochrone model. *arXiv:1411.4937*, 2014.

[54] C. M. Bender and S. A. Orszag. *Advanced Mathematical Methods for Scientists and Engineers I: Asymptotic Methods and Perturbation Theory*. Springer, New York, 1999.

[55] J. Barré and Y. Y. Yamaguchi. On algebraic damping close to inhomogeneous Vlasov equilibria in multi-dimensional spaces. *J. Phys. A: Math. Theor.*, 46:225501, 2013.

[56] M. D. Weinberg. Weakly damped modes in star clusters and galaxies. *Astrophys. J.*, 421:481–490, 1994.

[57] J.-B. Fouvry and S. Prunet. Linear response theory and damped modes of stellar clusters. *Mon. Not. R. Astron. Soc.*, 509:2443–2462, 2022.

[58] H. Yoshida. Construction of higher order symplectic integrators. *Phys. Lett. A*, 150:262–268, 1990.

[59] M. Gieles and A. Zocchi. A family of lowered isothermal models. *Mon. Not. R. Astron. Soc.*, 454:576–592, 2015.

[60] I. R. King. The structure of star clusters. III. Some simple dynamical models. *Astron. J.*, 71:64, 1966.

[61] C. P. Wilson. Dynamical models of elliptical galaxies. *Astron. J.*, 80:175, 1975.

[62] G. Strang. On the Construction and Comparison of Difference Schemes. *SIAM J. Numer. Anal.*, 5:506–517, 1968.

[63] Y. Hua and T. K. Sarkar. Matrix pencil method for estimating parameters of exponentially damped/undamped sinusoids in noise. *IEEE Trans. Acoust. Speech Signal Process.*, 38:814–824, 1990.

[64] NIST Digital Library of Mathematical Functions. `https://dlmf.nist.gov/`, Release 1.2.8 of 2026-09-15. F. W. J. Olver, A. B. Olde Daalhuis, D. W. Lozier, B. I. Schneider, R. F. Boisvert, C. W. Clark, B. R. Miller, B. V. Saunders, H. S. Cohl, and M. A. McClain, eds.

[65] P. D. Louis. Discrete oscillation modes and damped stationary density waves in one-dimensional collisionless systems. *Mon. Not. R. Astron. Soc.*, 258:552–570, 1992.

[66] T. Asano and T. Antoja. Delayed phase mixing in the self-gravitating Galactic disc. *Astron. Astrophys.*, 708:A226, 2026.

[67] I. Karpov, T. Argyropoulos, and E. Shaposhnikova. Thresholds for loss of Landau damping in longitudinal plane. *Phys. Rev. Accel. Beams*, 24:011002, 2021.

[68] T. O'Neil. Collisionless damping of nonlinear plasma oscillations. *Phys. Fluids*, 8:2255–2262, 1965.

[69] P. D. Louis and O. E. Gerhard. Can galaxies oscillate? A self-consistent model of a non-stationary stellar system. *Mon. Not. R. Astron. Soc.*, 233:337–365, 1988.